HYPERSPECTRAL & POLARIZATION IMAGING

Computational Imaging – Spring 2024

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Previously on
COMPUTATIONAL IMAGING

Plenoptic imaging

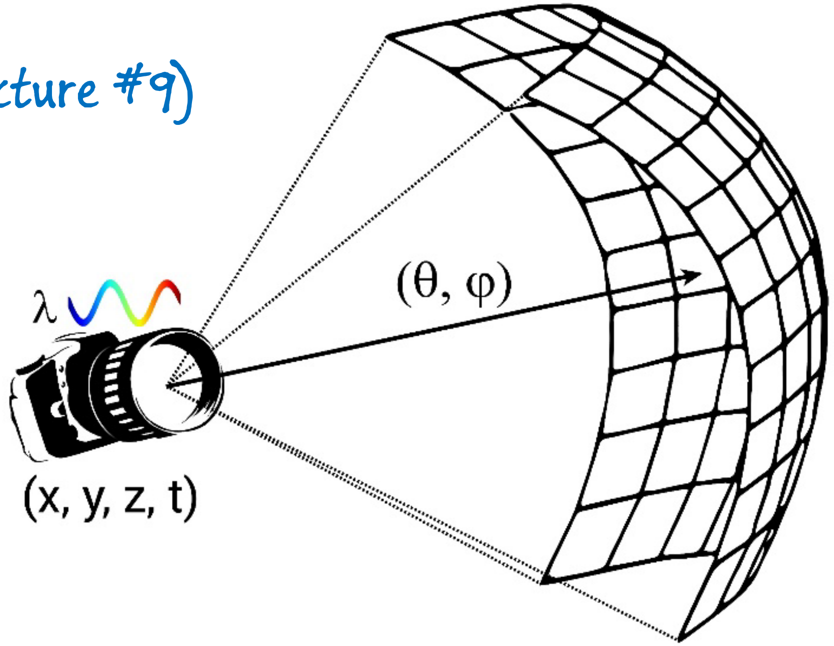
Basis:

- Digital photography (Lecture #1)
- Mathematical foundations (Lecture #2)

Transient imaging (Lecture #9)

Hyperspectral imaging (Lecture #8)

$$P = P(\theta, \underline{\phi}, \lambda, t, \underline{x, y, z})$$



Light field imaging (Lecture #4)

High dynamic range imaging (Lecture #5)

Computational displays (Lecture #6)

Plus common tools and techniques:

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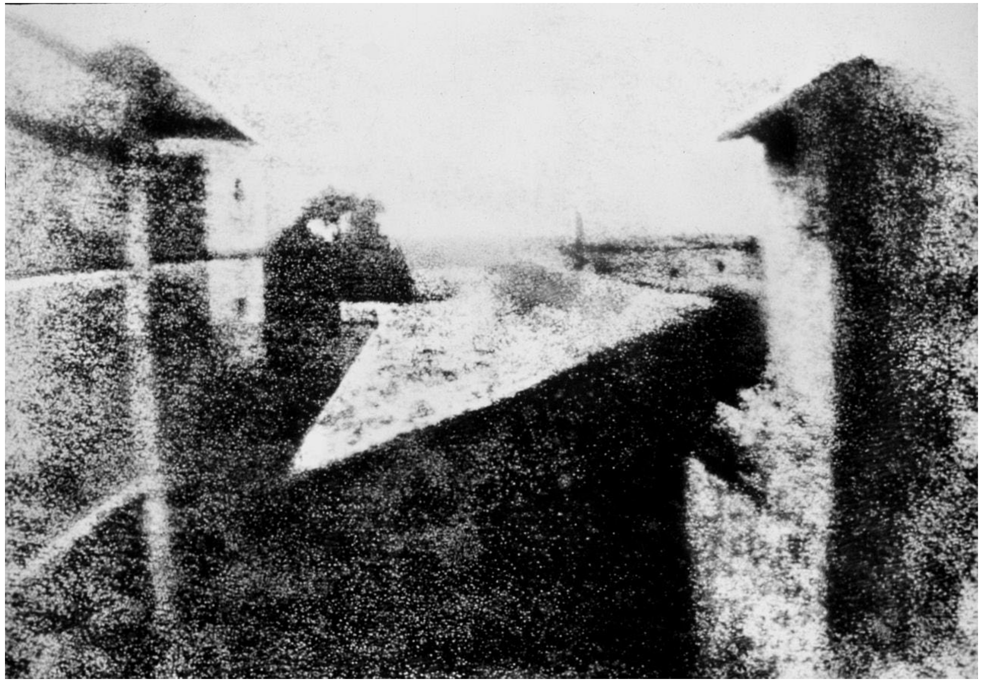
- Compressive sensing (Lecture #3)
- Computational illumination (Lecture #7)

What is an image?

An image is a 2D realization of a 3D world:



View from the Window at Le Gras (1826)

What is an image?

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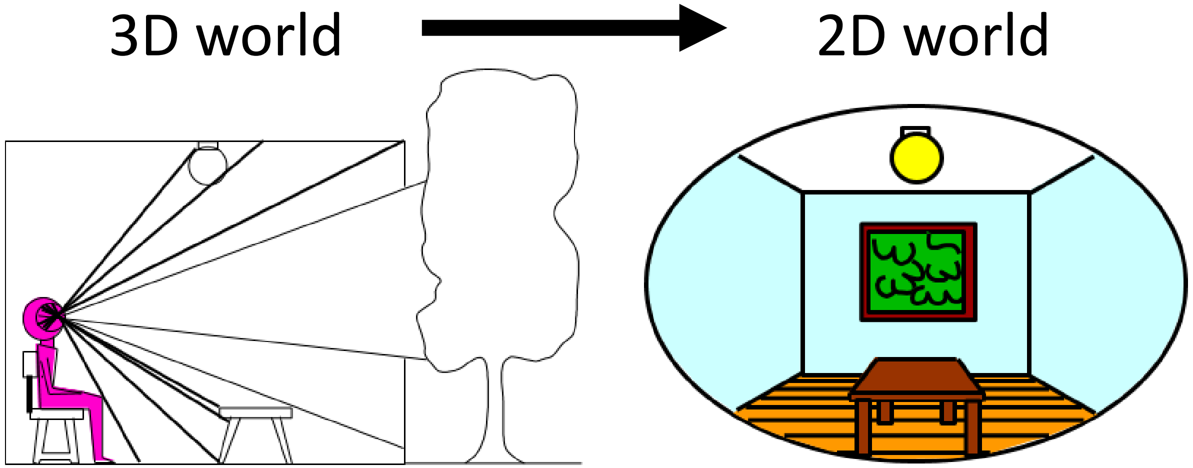
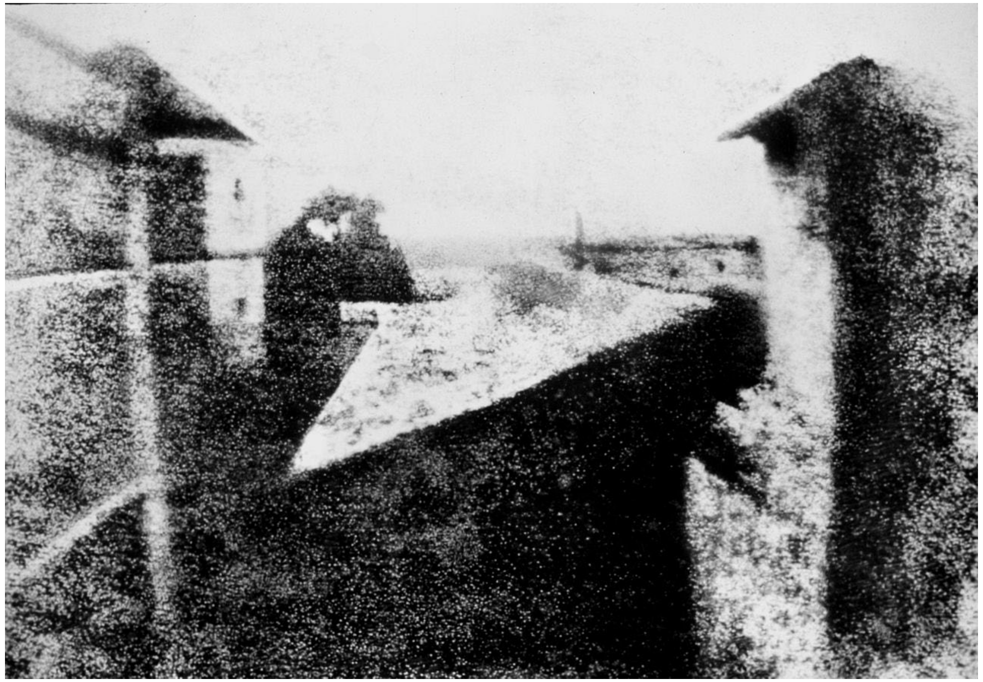


What do we lose?



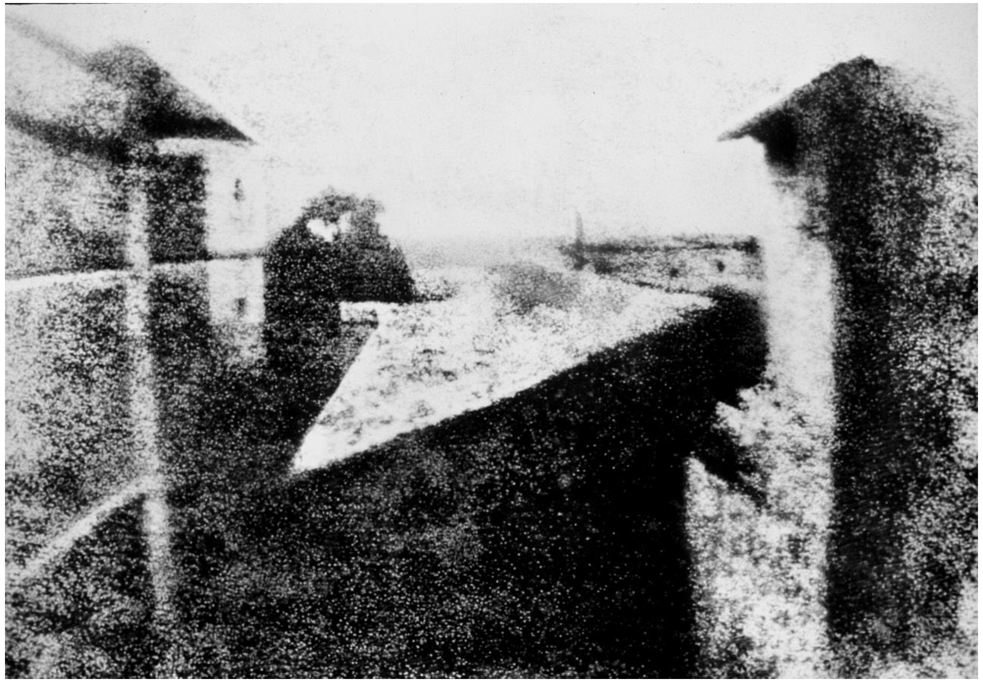
View from the Window at Le Gras (1826)

What is an image?

An image is a 2D realization of a 3D world:

- Usually *depicts* visual perception ***
- Usually *simplifies* the physical world ***
- Usually *represents* visible light, color ***

But not always!



View from the Window at Le Gras (1826)

What is an image?

Mathematically speaking:

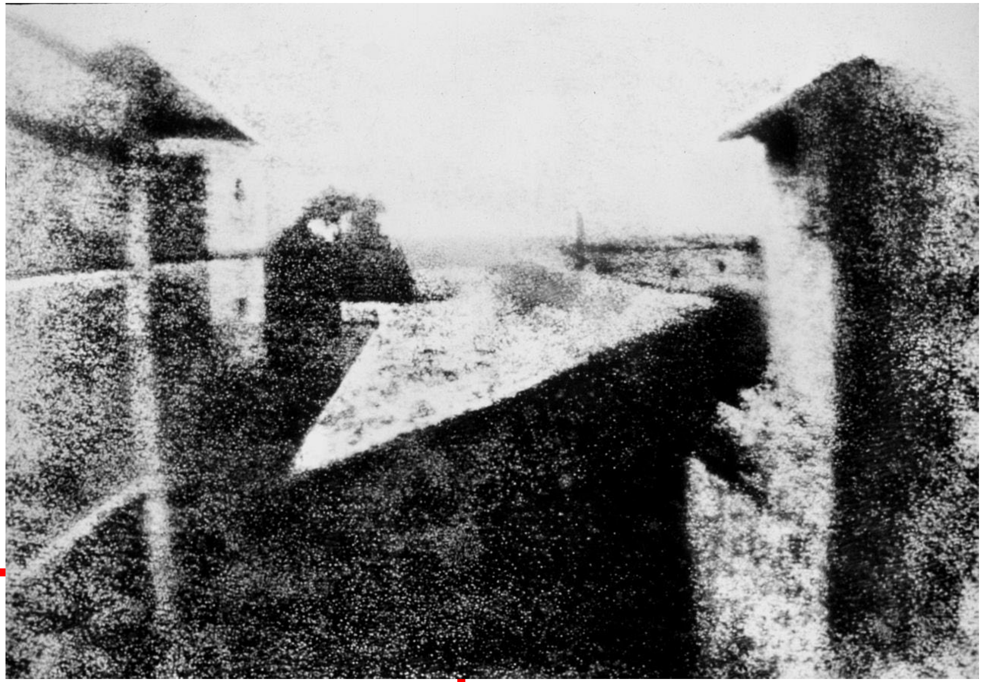
$$I : \mathbb{R}^2 \longmapsto color$$

How do we generate I from a 3D world?

- Eye? Camera? Other device?
- How much definition? How “wide” can we see?

How do we capture and encode color?

- Digitally? Analogically? Human eye?
- B/W images: Single color value (i.e. intensity)?
- Color values: Tristimuli? More accurate? Intensity range?



View from the Window at Le Gras (1826)

LIGHT 101

Inspired by a slide by Fredo Durand

What is light?

- Arguably, one of the greatest philosophical questions of all time.
  - Empedocles (s. V b.C.), Plato (s.IV b.C.), Aristotle (s.IV b.C.)
    - *Internal emitted fire connected with the eye* ***(Emission hypothesis)***
  - Lucrecius (s. I b.C.) –Newton (s. XVII-XVIII a.D.)
    - *Little particles traveling very, very fast* ***(Particle hypothesis)***
  - Descartes, Huygens, Fresnel, Young (s. XVII)
    - *A wave propagating through the ether* ***(Wave hypothesis)***
  - Gauss, Faraday, Maxwell (s.XVIII-XIX)
    - *An electromagnetic wave* ***(EM hypothesis)***
  - Plank et al. (s.XIX-XX)
    - *A quantized electromagnetic wave* ***(Quantum hypothesis)***

What is light?

$$\nabla \cdot \mathbf{D} = \rho$$

$$\nabla \cdot \mathbf{B} = 0$$

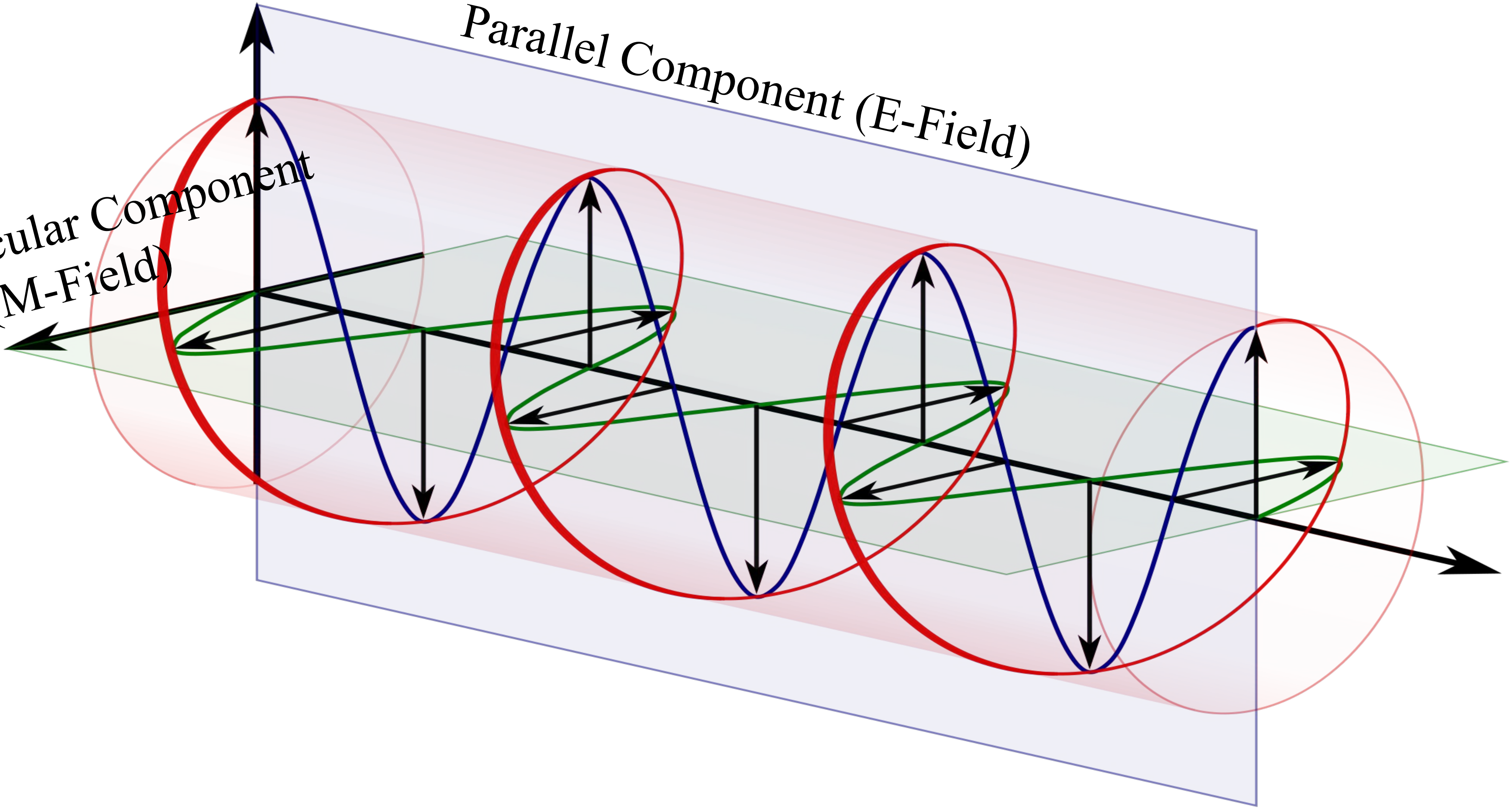
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

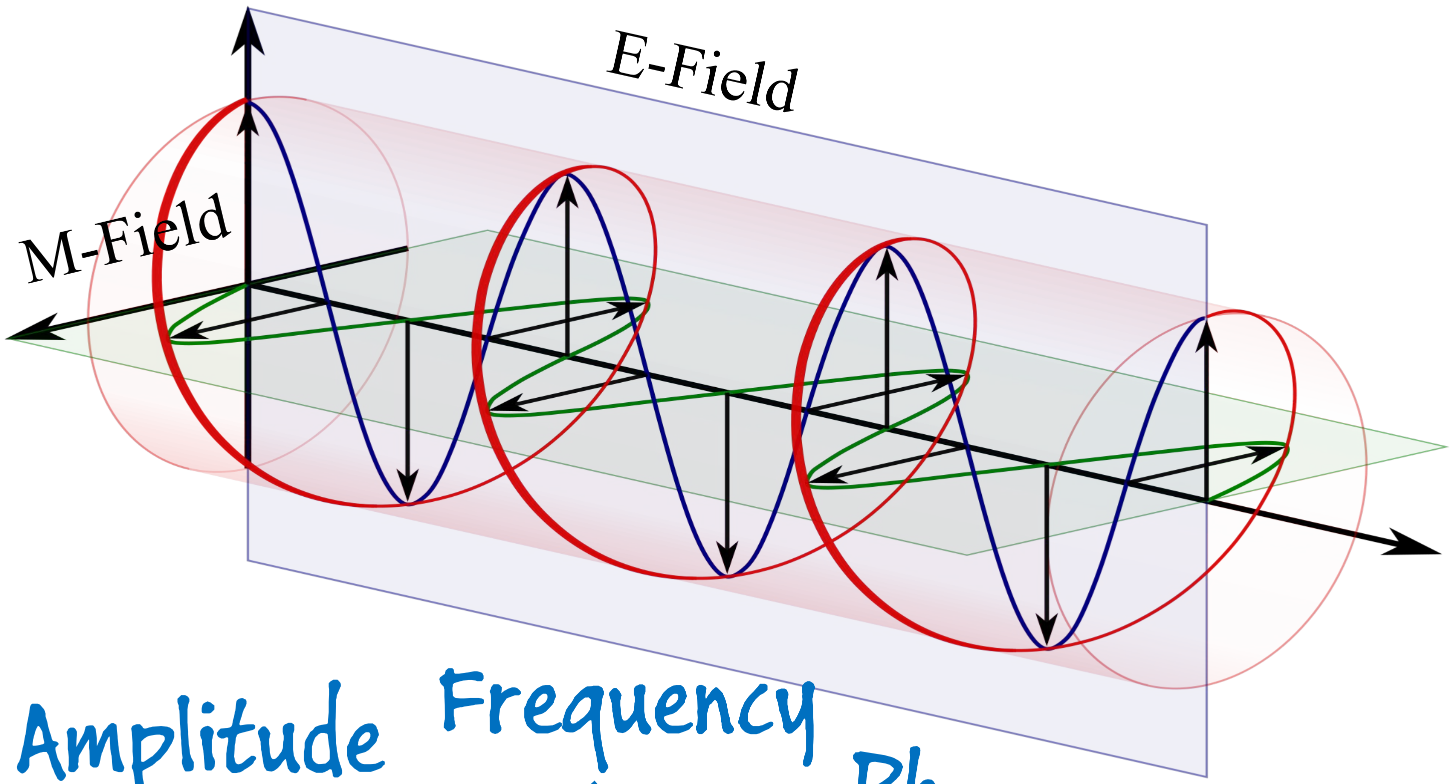
And assuming a time-harmonic monochromatic planar field

James Clerk Maxwell (1831-1979)

Light as an electromagnetic wave



Light as an electromagnetic wave

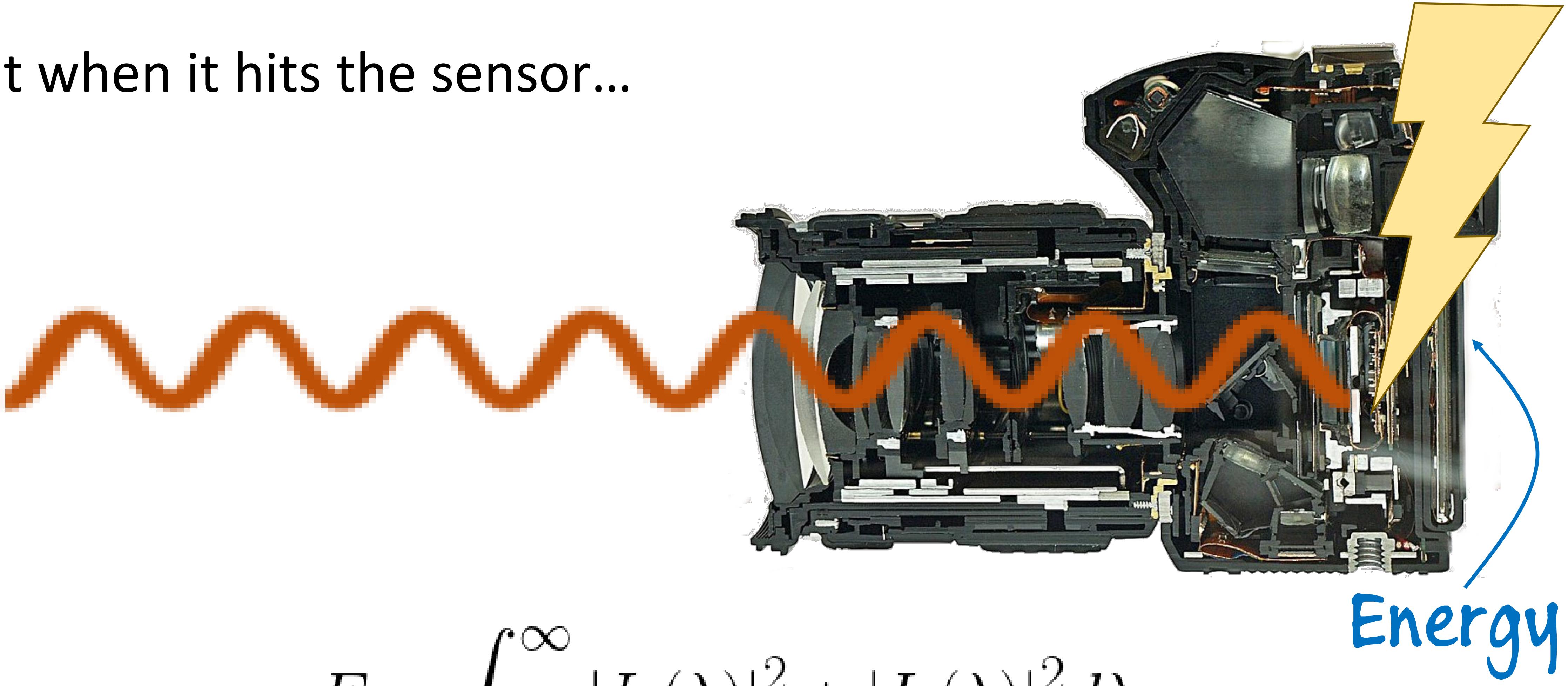


Amplitude Frequency Phase

$$E_{\mathbf{xy}}(r) = \langle I_s \sin(\omega r + \phi_s), I_p \sin(\omega r + \phi_p) \rangle$$

Light as an electromagnetic wave

- But when it hits the sensor…



$$E = \int_0^\infty |I_p(\lambda)|^2 + |I_s(\lambda)|^2 d\lambda$$

In short… what do we capture/ what do we lose?

$$E = \int_0^{\infty} |I_p(\lambda)|^2 + |I_s(\lambda)|^2 d\lambda$$

- Cameras capture intensity (plain and simple)
- We filter this intensity to give more weight to certain wavelengths
  - Hoping to resemble human perception

- So we forever lose…
  - The **spectrally-resolved intensity**
  - The components of each phasor: **Amplitude and phase**

Plenoptic imaging

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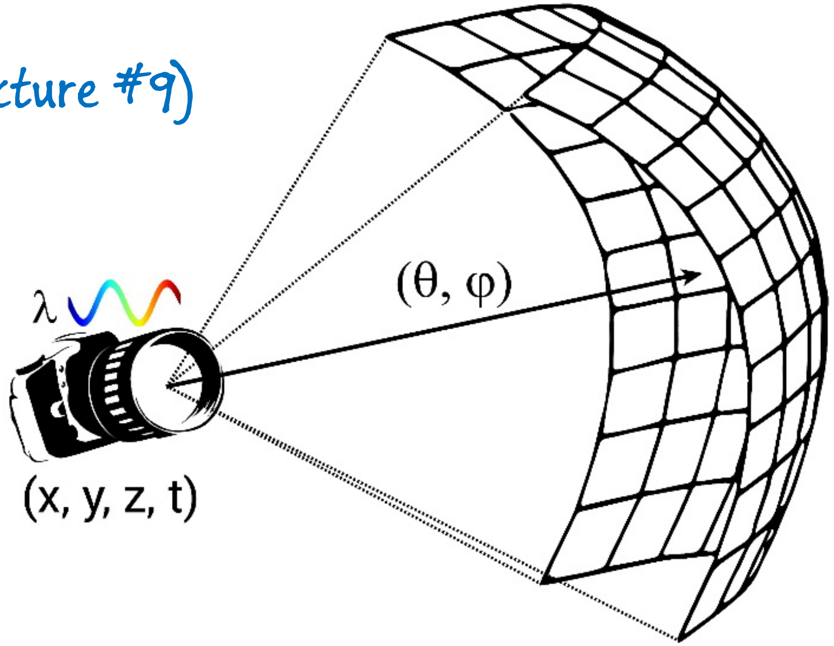

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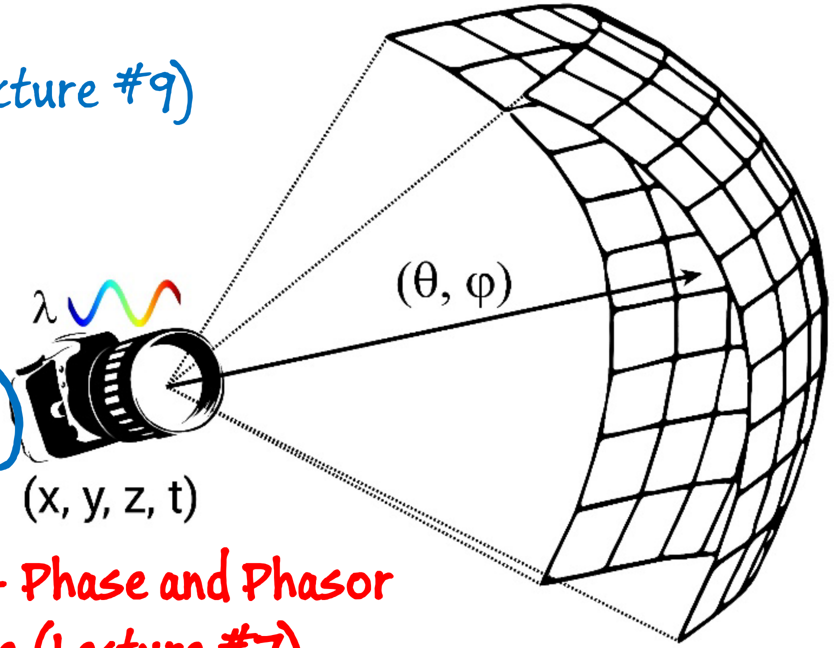

Light field imaging (Lecture #4)

Polarization - Phase and Phasor Amplitude (Lecture #7)

High dynamic range imaging (Lecture #5)

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Plus common tools and techniques:

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Extended Plenoptic imaging

Hyperspectral imaging (Lecture #8)

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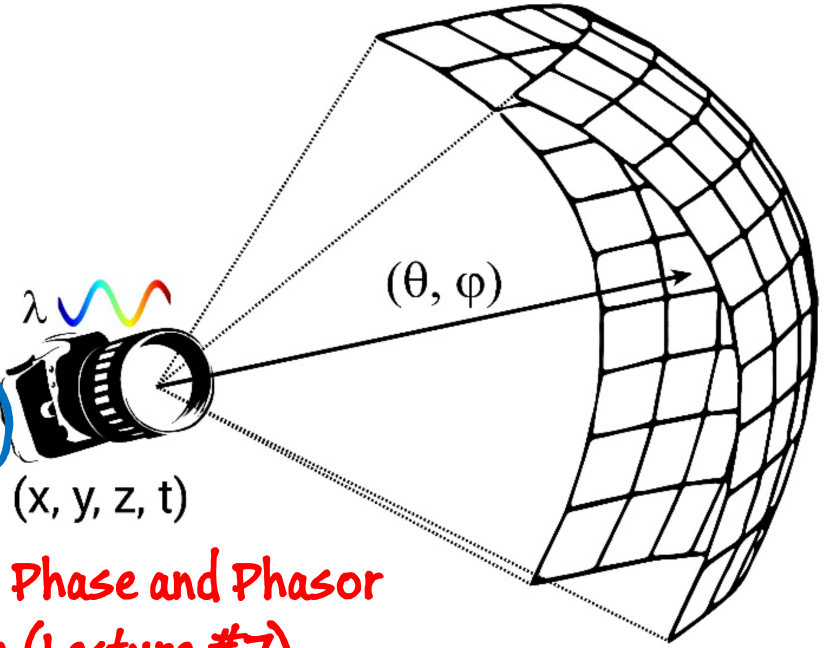


Polarization - Phase and Phasor Amplitude (Lecture #7)

- Why are these interesting/important?
- How can we capture them back with our camera.

Questions?

James Clerk Maxwell (1831-1979)

HYPERSPECTRAL IMAGING

Source: Fredo Durand

Hyperspectral Imaging

- **What** is it?

- **Why** is it important? (a.k.a. applications)

- **How** do we capture it?

Color

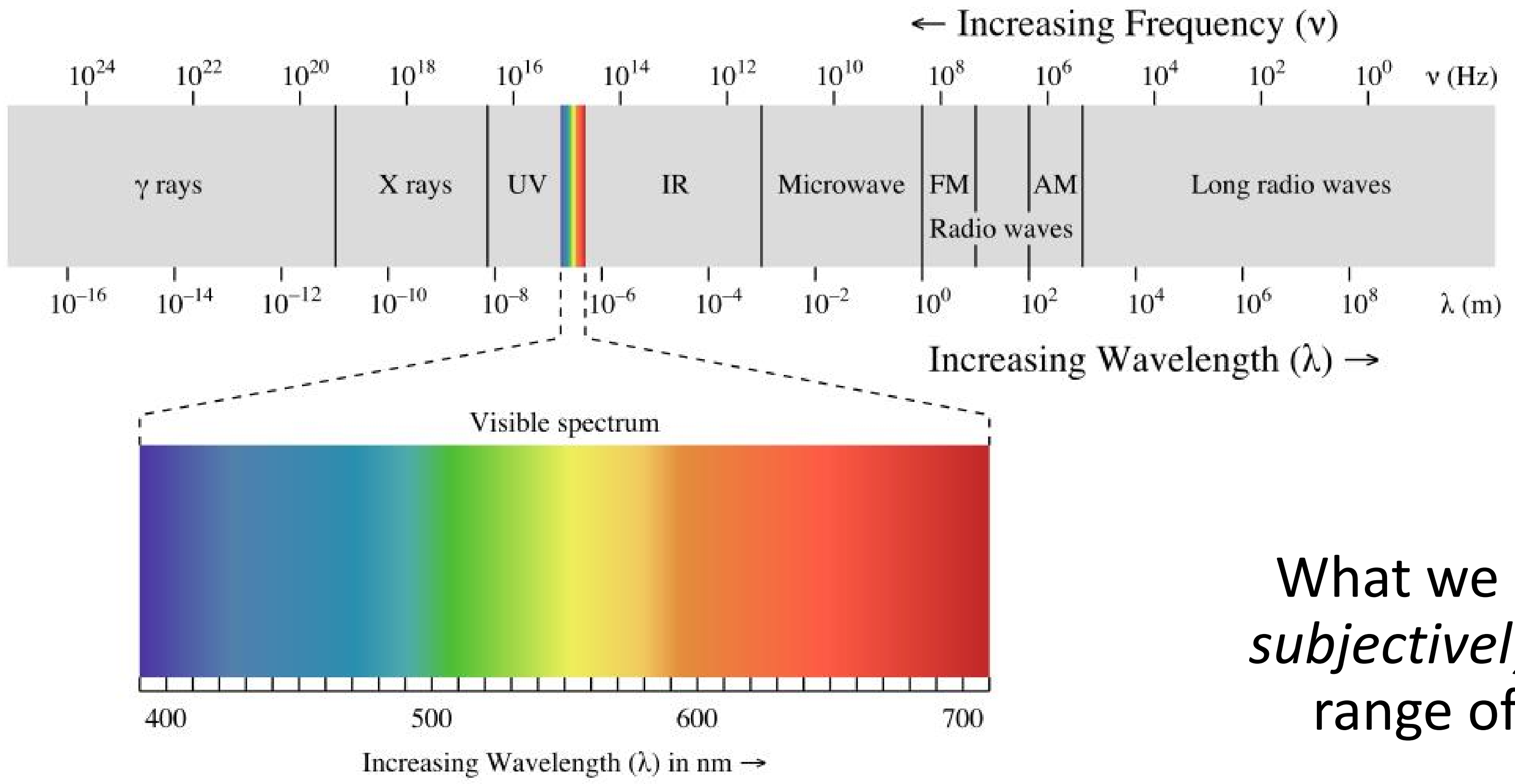
- Very high-level of color as it relates to digital photography.

- We could spend an entire course covering color.

color is complicated

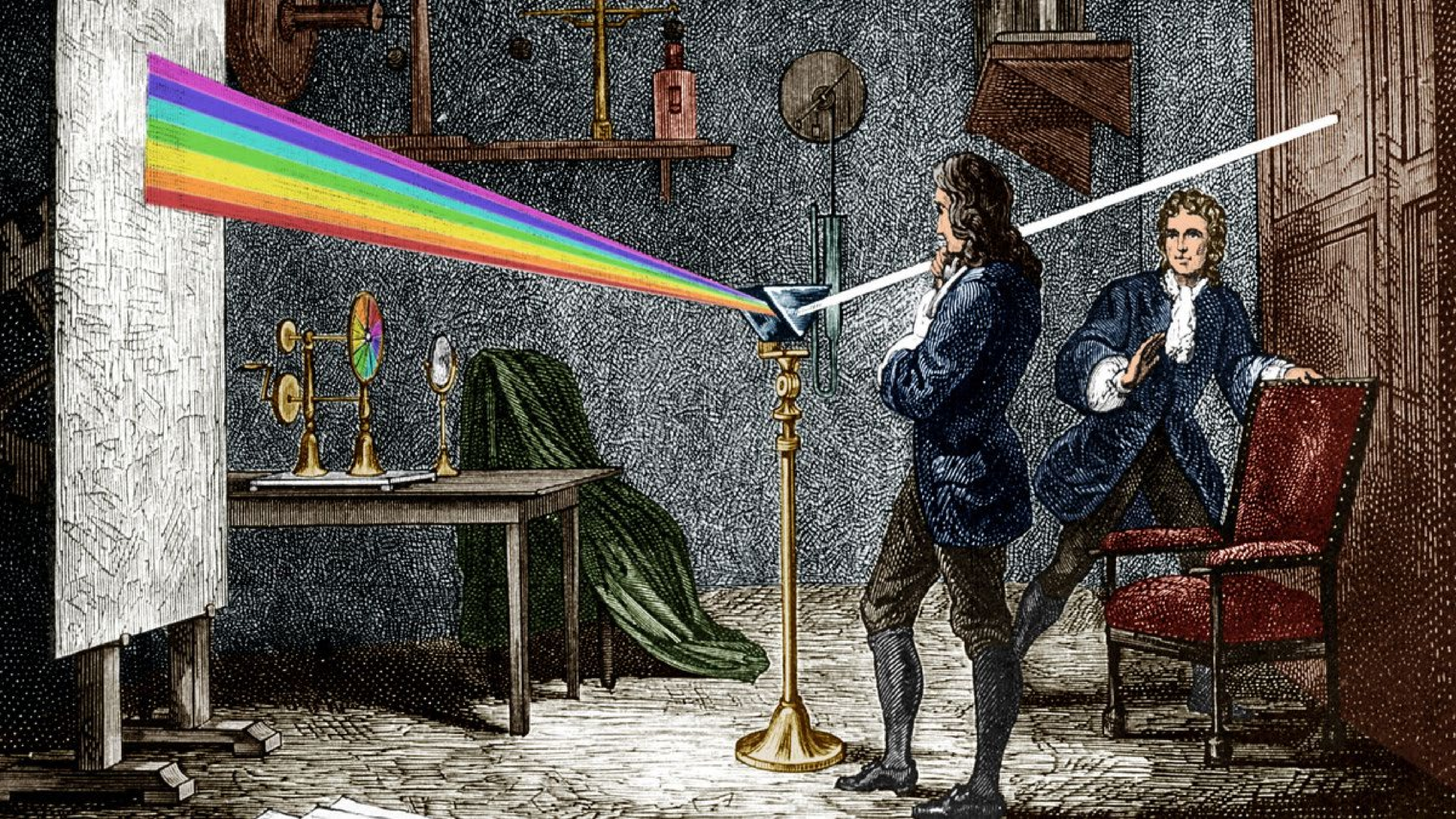
Color is an artifact of human perception

- “Color” is not an *objective* physical property of light (electromagnetic radiation).
- Instead, light is characterized by its wavelength.

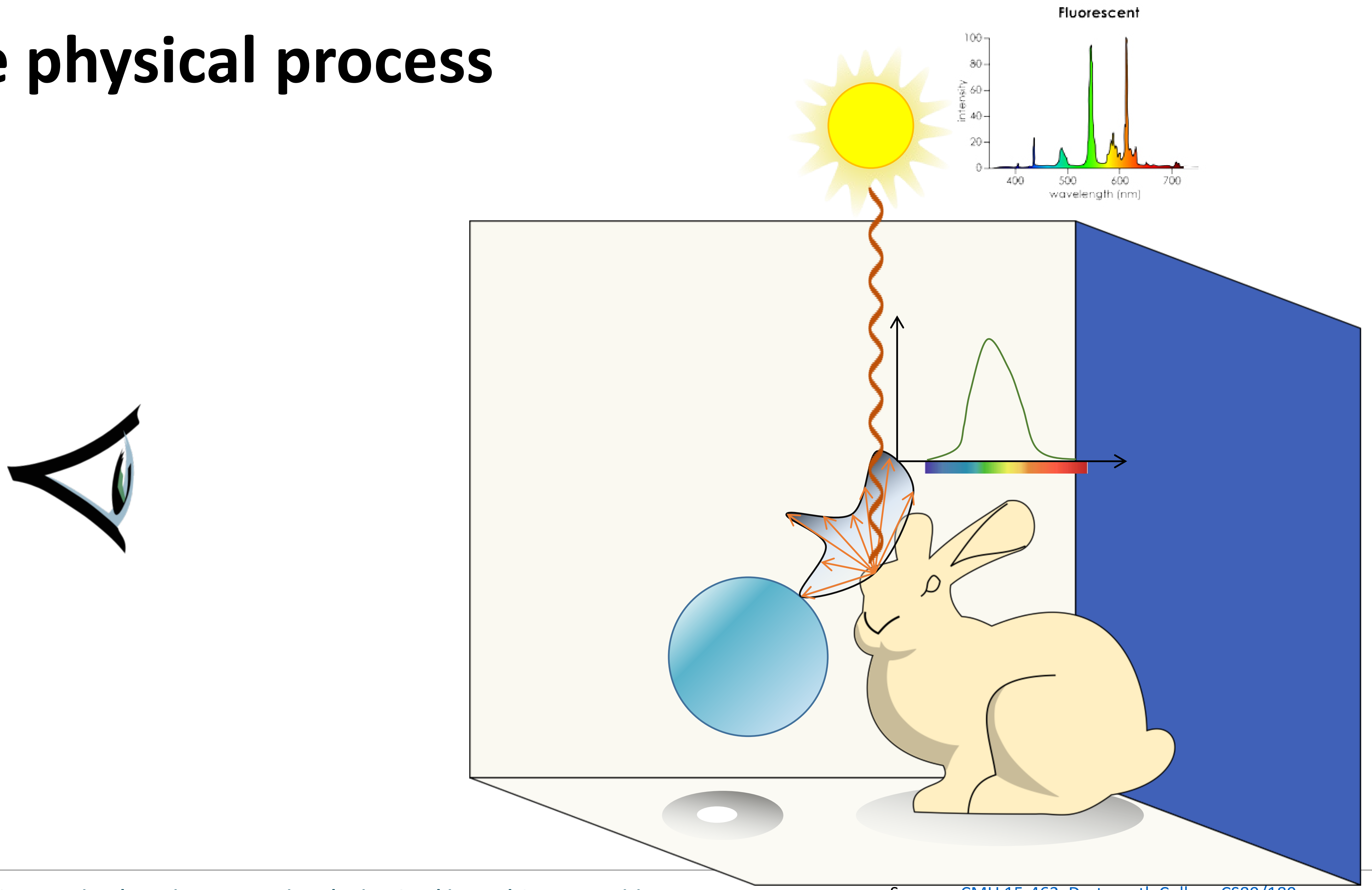


electromagnetic spectrum

What we call “color” is how we *subjectively* perceive a very small range of these wavelengths.

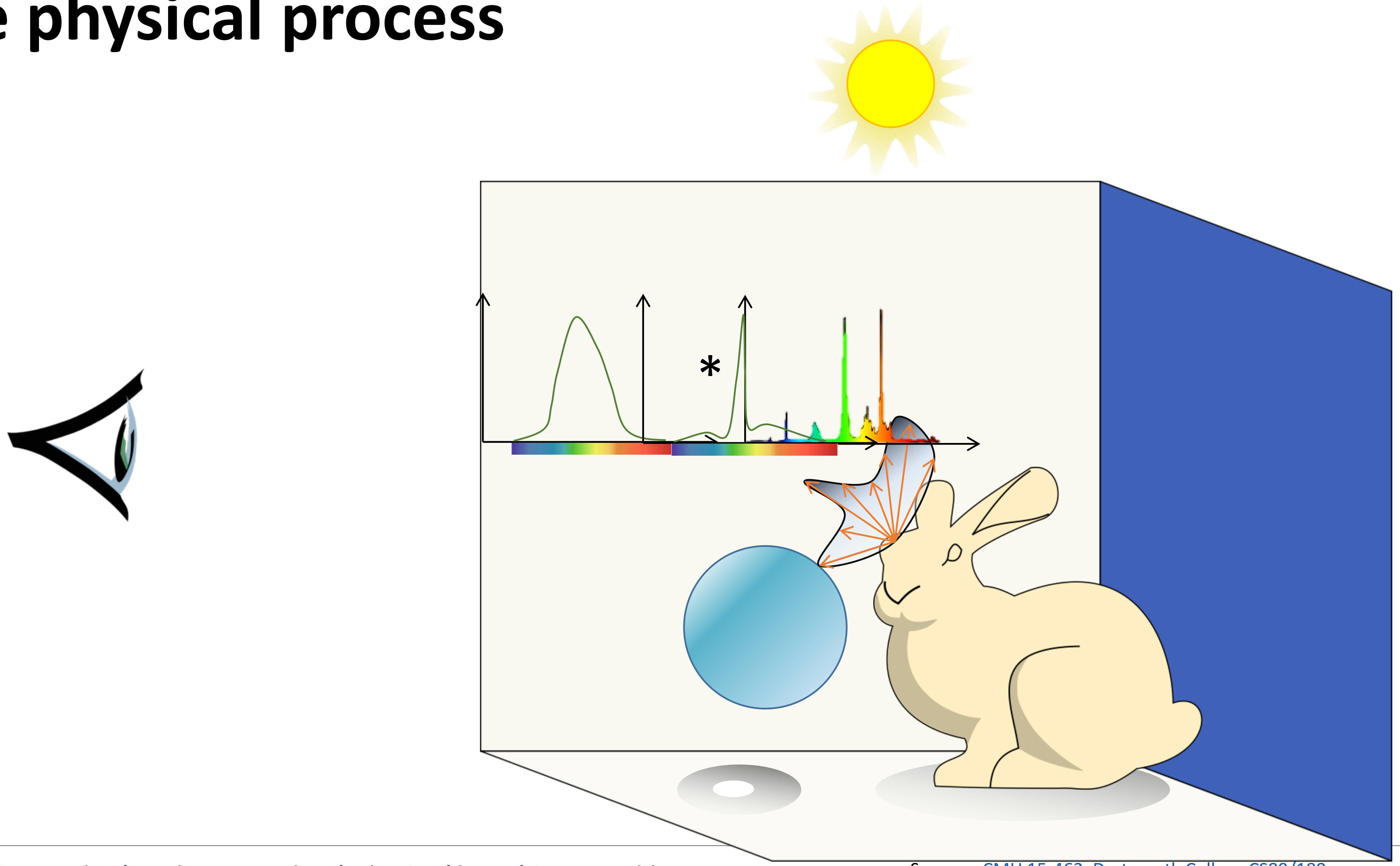


The physical process



Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

The physical process



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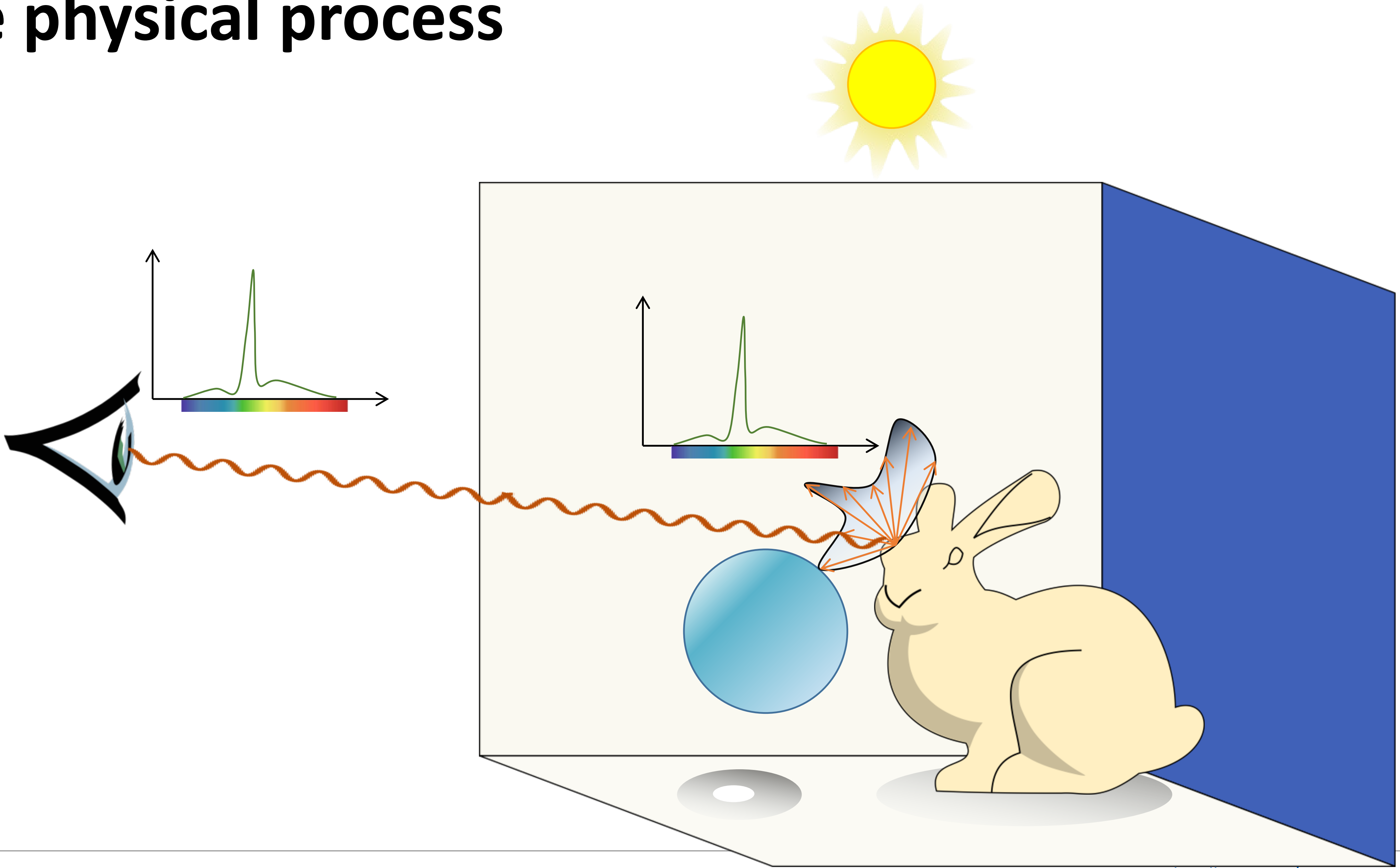

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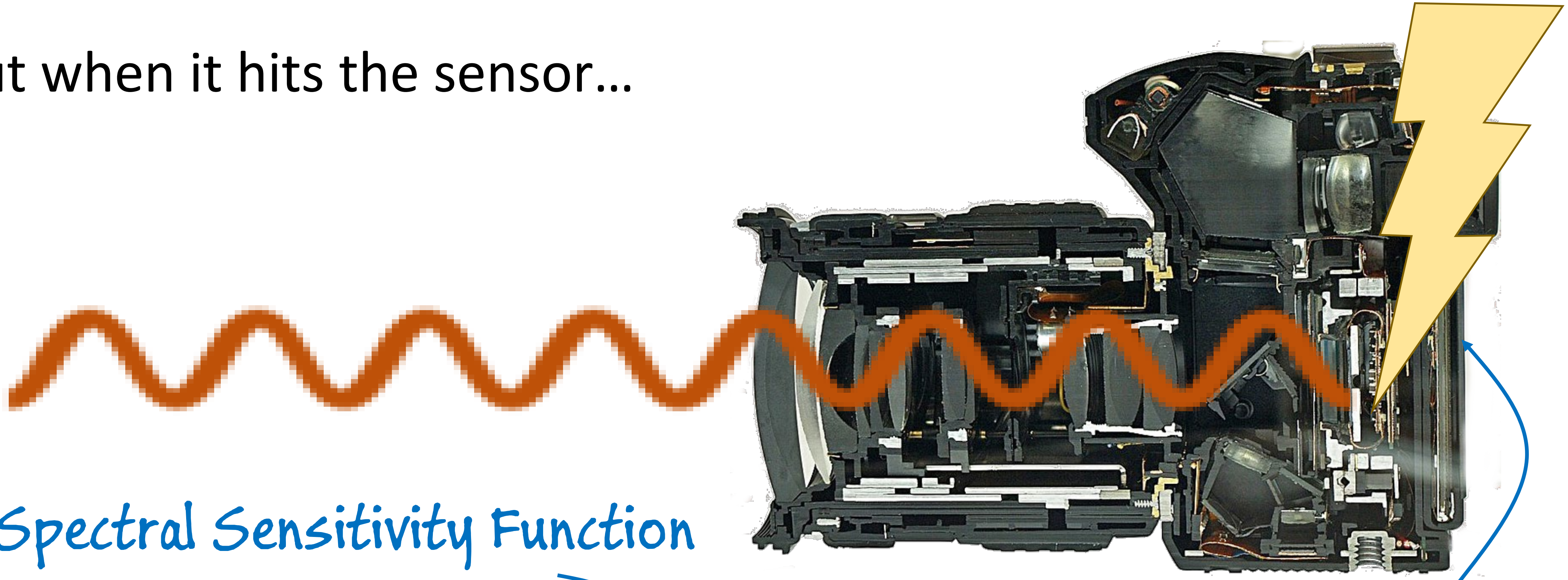
Light as an electromagnetic wave

- But when it hits the sensor…



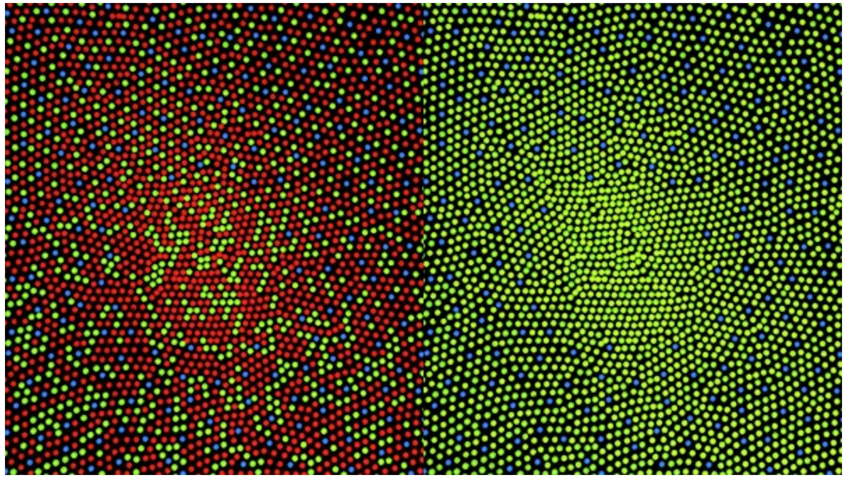
Spectral Sensitivity Function

Energy

$$E = \int_0^\infty S(\lambda)\left(|I_p(\lambda)|^2 + |I_s(\lambda)|^2\right) d\lambda$$

Remember – SSF of Human Eyes

- The human eye is a collection of light sensors called cone cells.
- There are three types of cells with different spectral sensitivity functions.
- Human color perception is three-dimensional (*tristimulus color*).

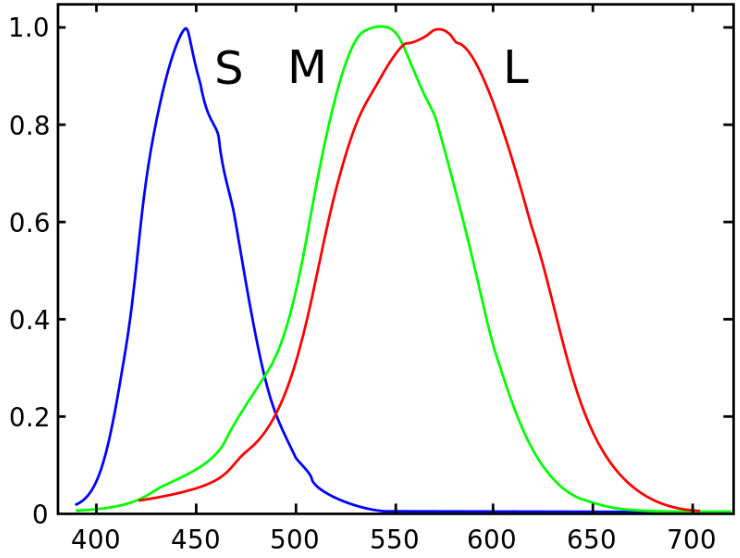


cone distribution for normal vision
(64% L, 32% M)

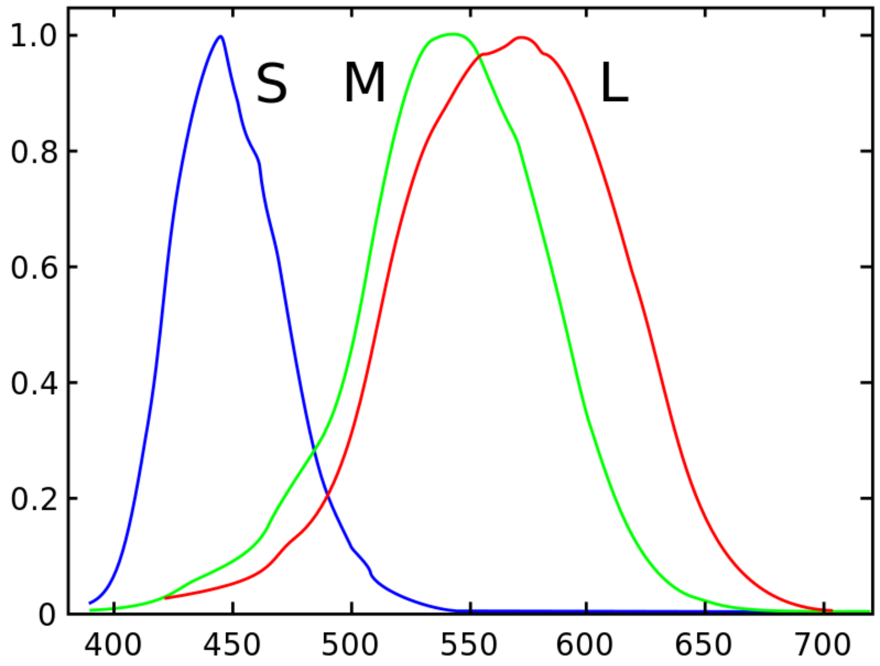
"short" $S = \int_\lambda \Phi(\lambda) S(\lambda) d\lambda$

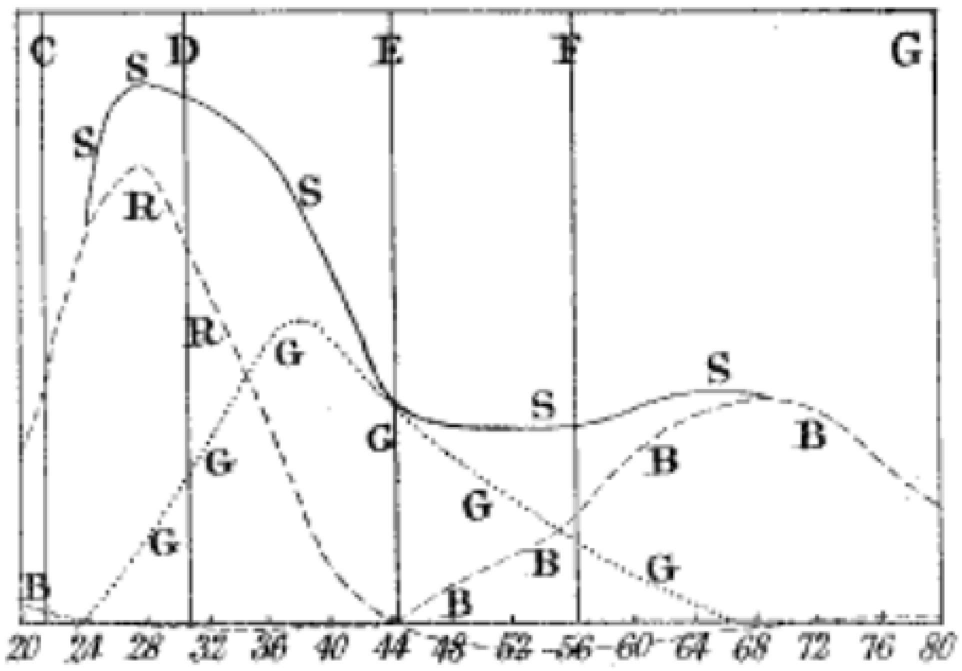
"medium" $M = \int_\lambda \Phi(\lambda) M(\lambda) d\lambda$

"long" $L = \int_\lambda \Phi(\lambda) L(\lambda) d\lambda$



Remember – SSF of Human Eyes





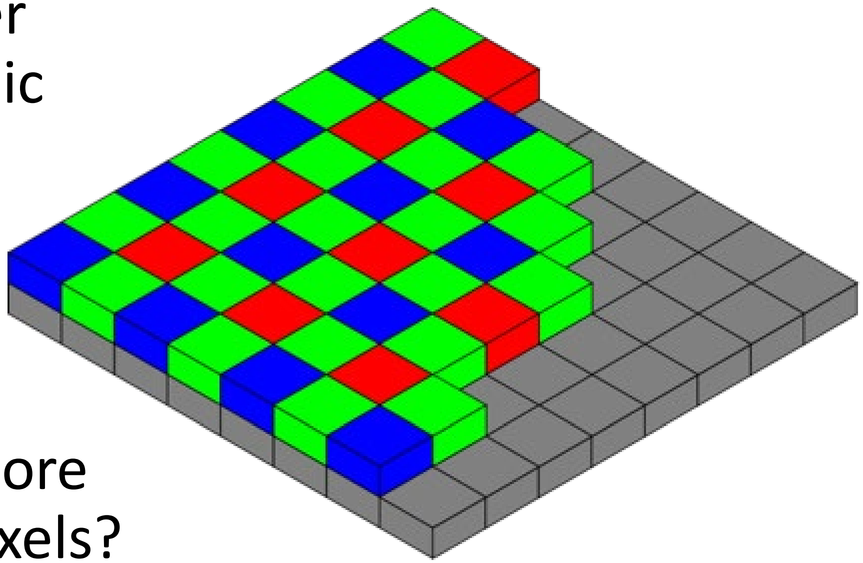
James Clerk Maxwell, 1860. *"On the Theory of Compound Colours, and the Relations of the Colours of the Spectrum."* Philosophical Transactions of the Royal Society of London

Remember – SSF of Digital Cameras

Two design choices:

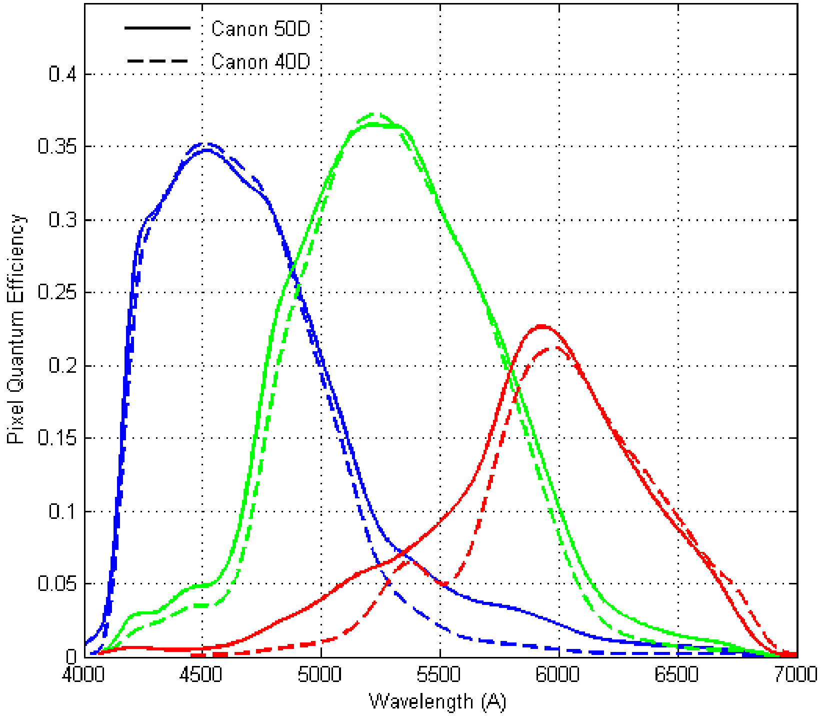
- What spectral sensitivity functions $f(\lambda)$ to use for each color filter?
- How to spatially arrange ("mosaic") different color filters?

Bayer mosaic



Why more green pixels?

SSF for Canon 50D



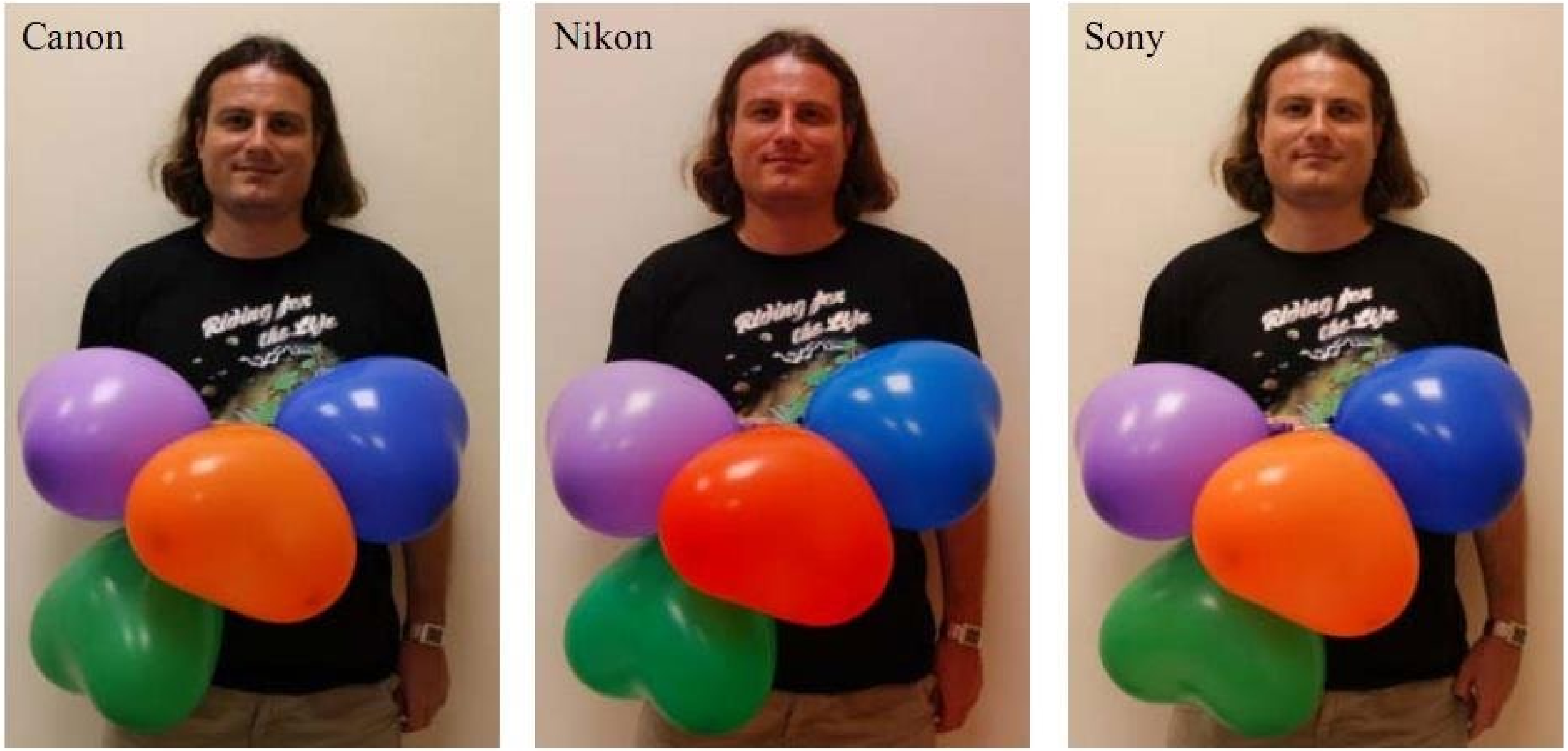
Generally do not match human LMS.

$f(\lambda)$

Remember – Each Dig. Camera has its own SSF

Each camera has its more or less unique, and most of the time *secret*, SSF.

- Makes it very difficult to correctly reproduce the color of sensor measurements.



So…

- Cameras (and our eyes) do not capture the electromagnetic spectrum

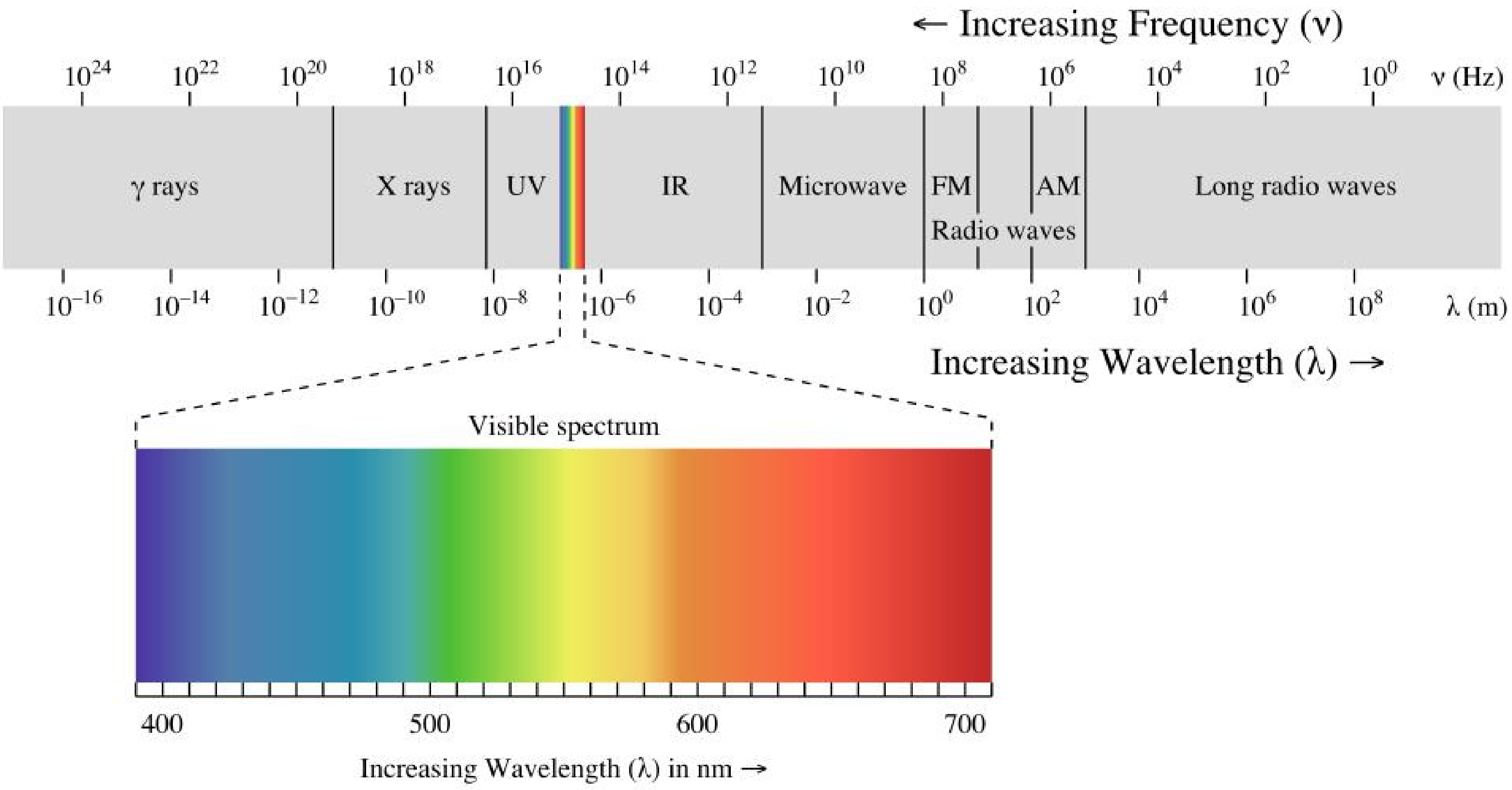
- As best, they coarsely sample the spectrum by filtering out frequencies
  - Not even that…

- (Very) incomplete description of the EM field/reflectance.
  - Spectroradiometry != Color

Hyperspectral Imaging

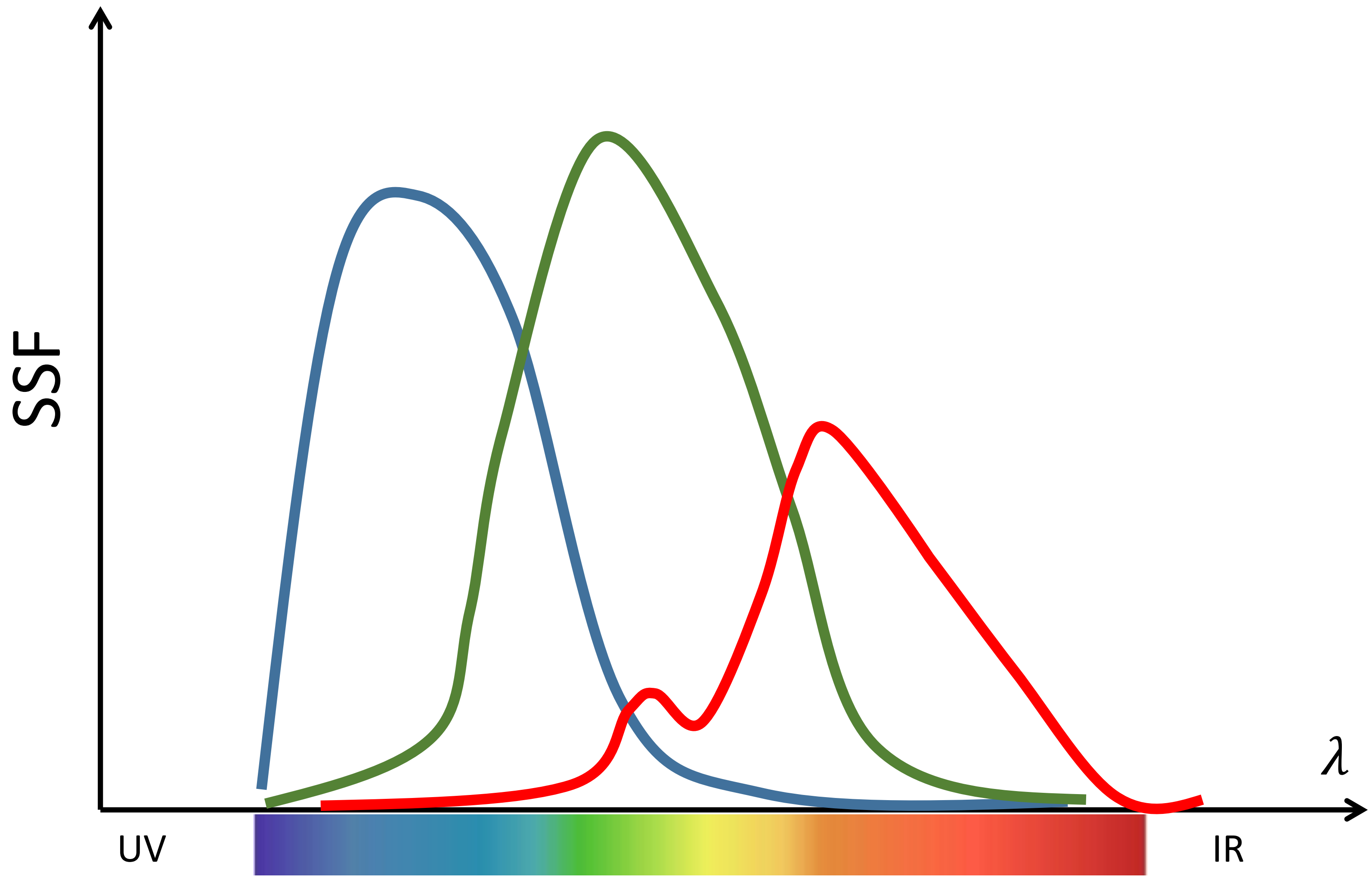
- Cameras (and eyes) do not capture the electromagnetic range
- … in fact, little-to-none sensitivity out of the visible spectrum
- **Hyperspectral imaging** tries to:
    a) Sample more finely the EM spectrum
    b) Recover (part of) the frequencies outside the visible

EM Spectrum

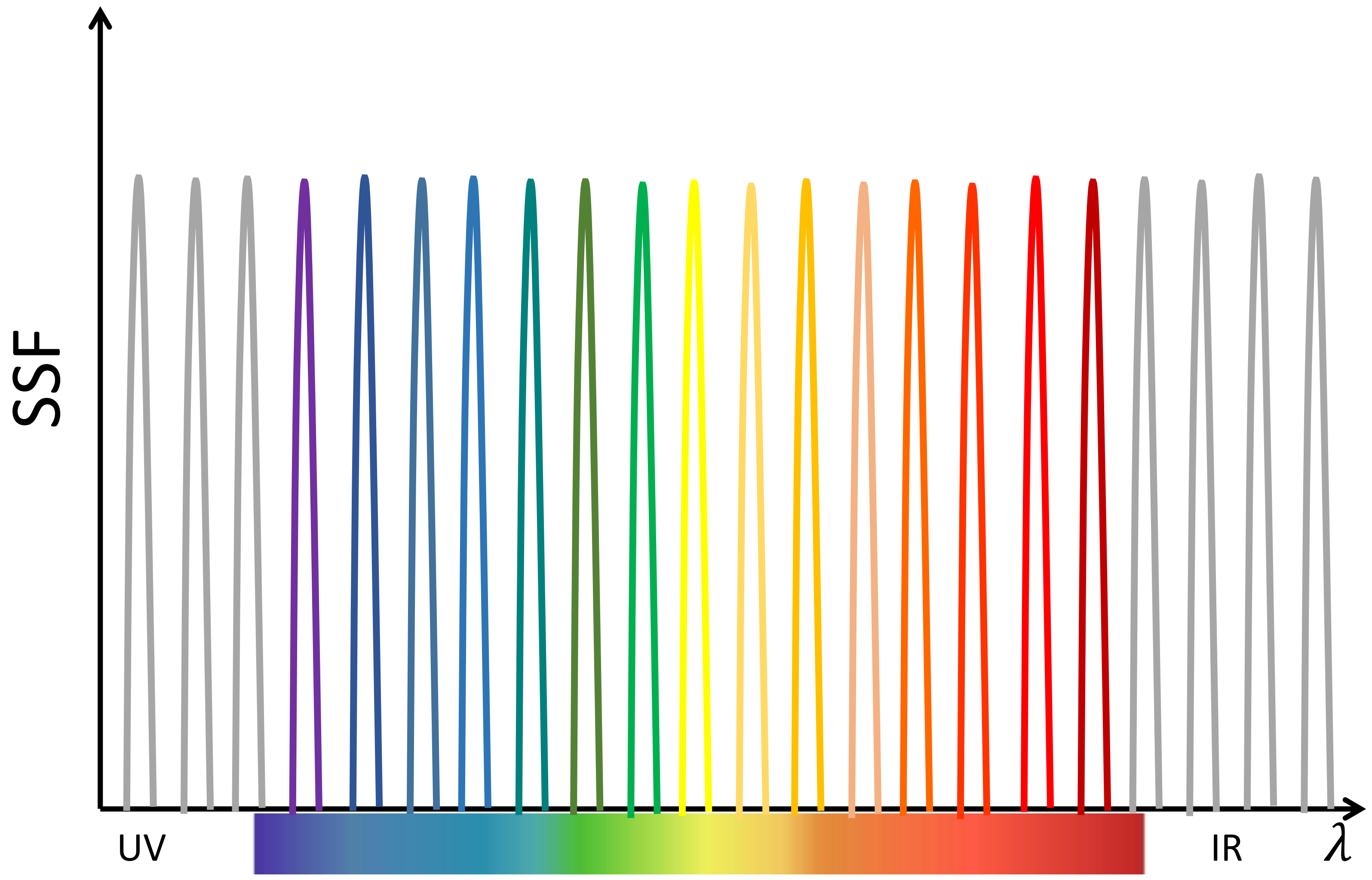


electromagnetic
spectrum

EM Spectrum



EM Spectrum



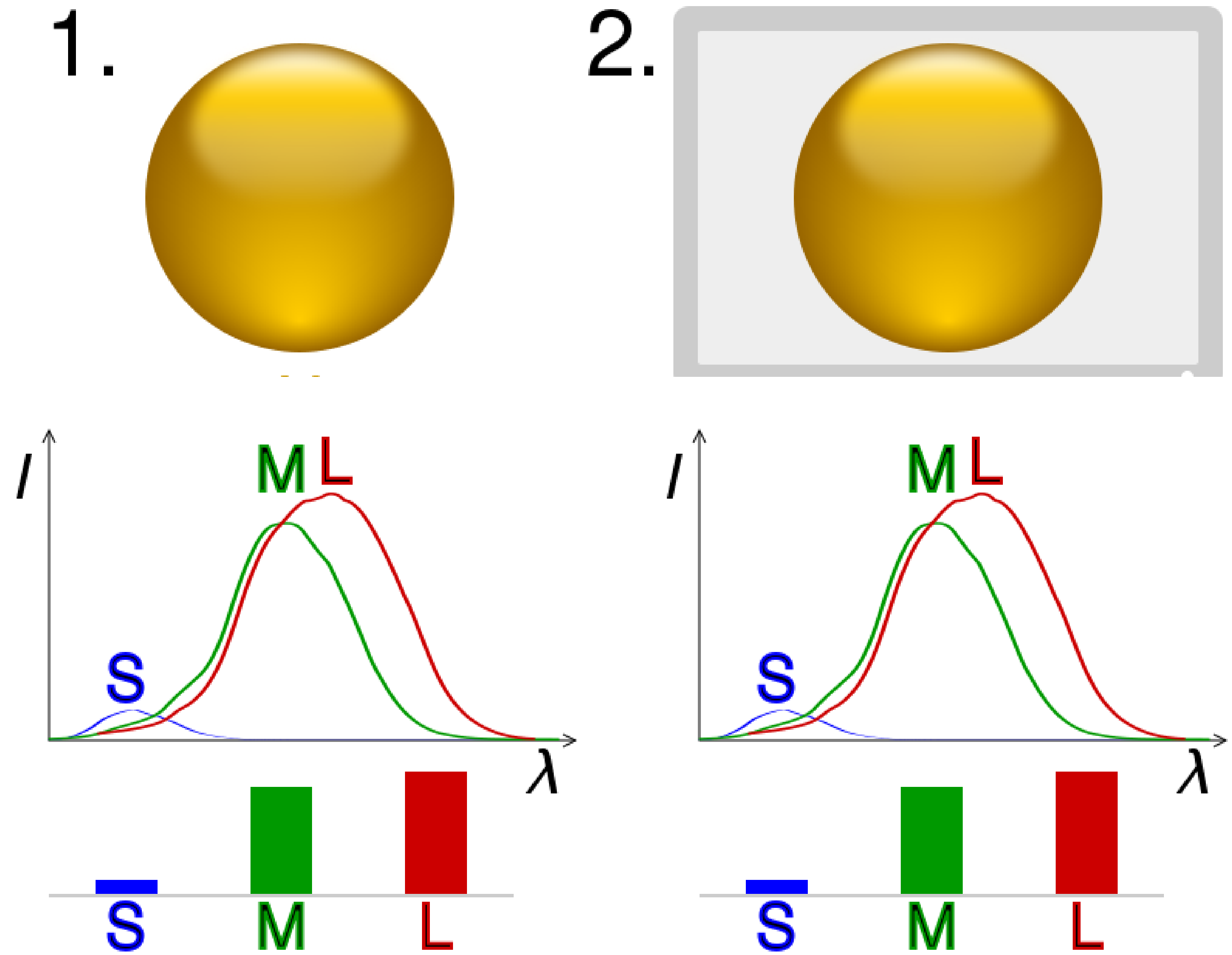
Why do we want to capture the full spectrum?

- We have seen that we are missing lots of (visual) info about the EM spectrum…

- … but how important is that info?
- Remember that RGB is optimized to our visual system

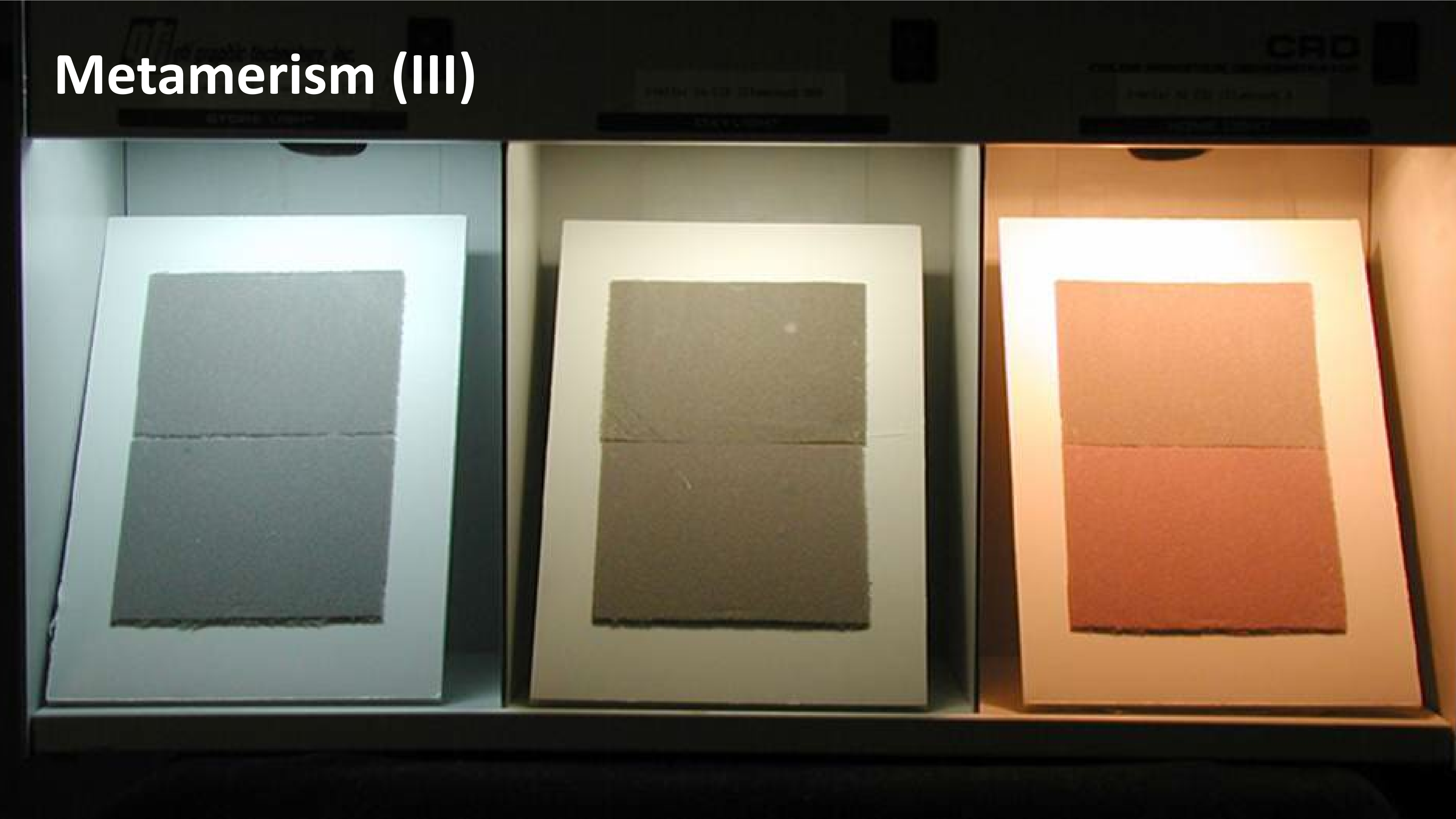
- **Brainstorming:** Where would we like to have HS info?

Metamerism (I)



Metamerism (II)

Metamerism (III)



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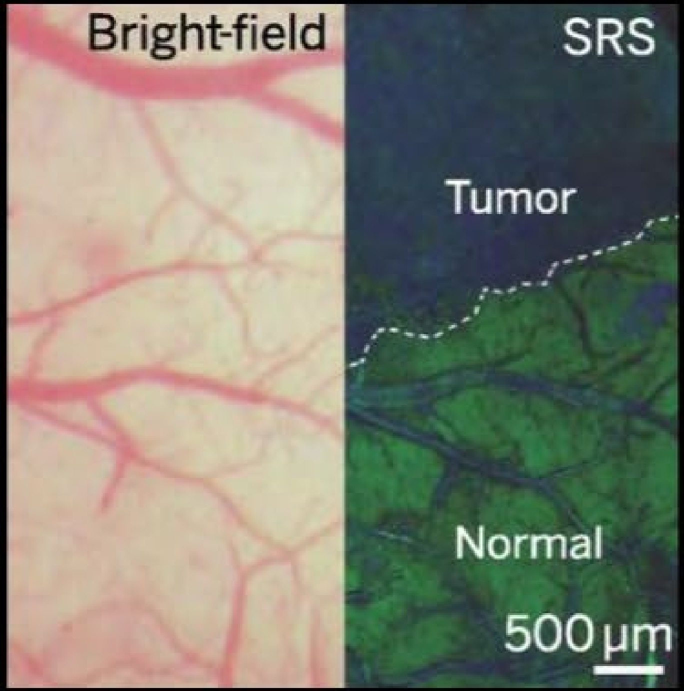
Hyperspectral Imaging – *Where is it important?*

Geology [1]

Cosmetics [2]



Biology [3]

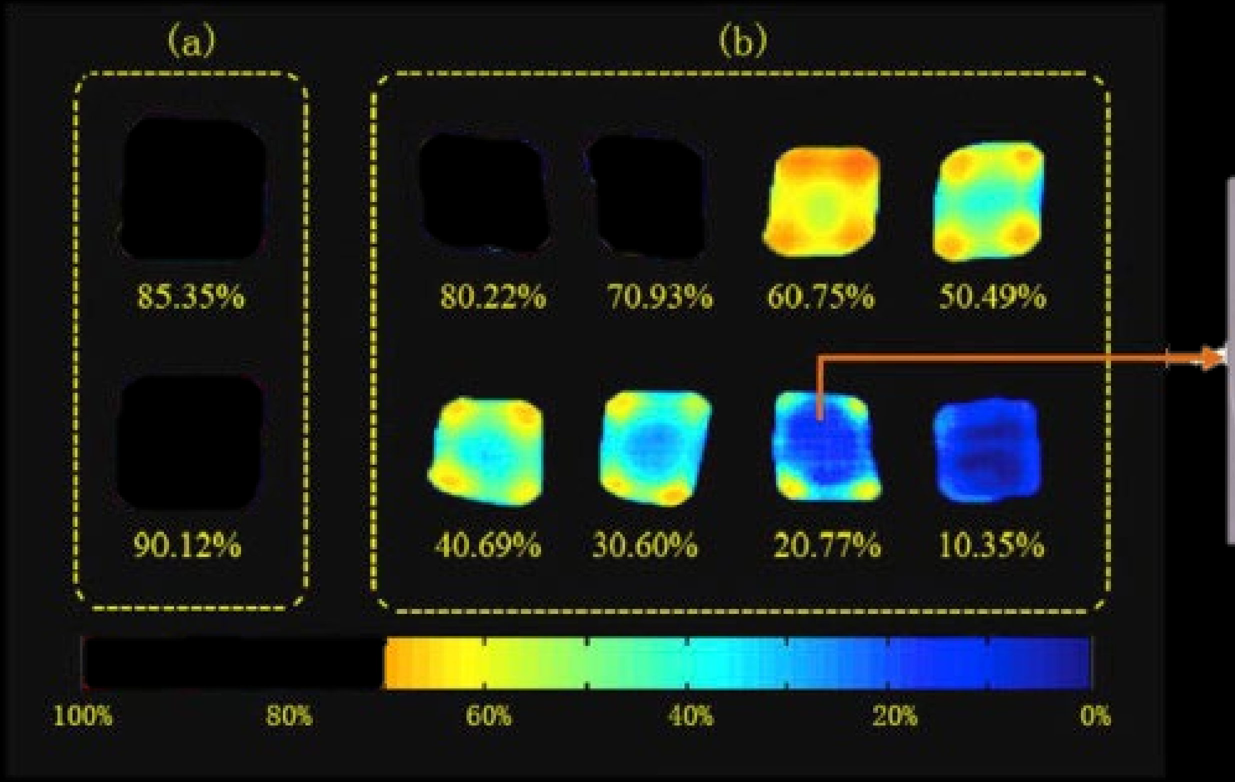
[1] NASA
[3] rshephorse, Flickr
[2] Cheng et al., Vibrational spectroscopic imaging of living systems: An emerging platform for biology and medicine, Science 2015

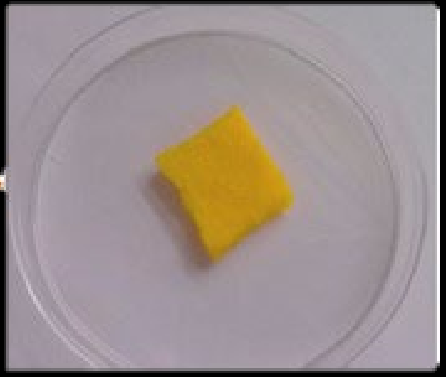
5

Source: Baek et al. 2017

Hyperspectral Imaging – *Where is it important?*

Quality Control

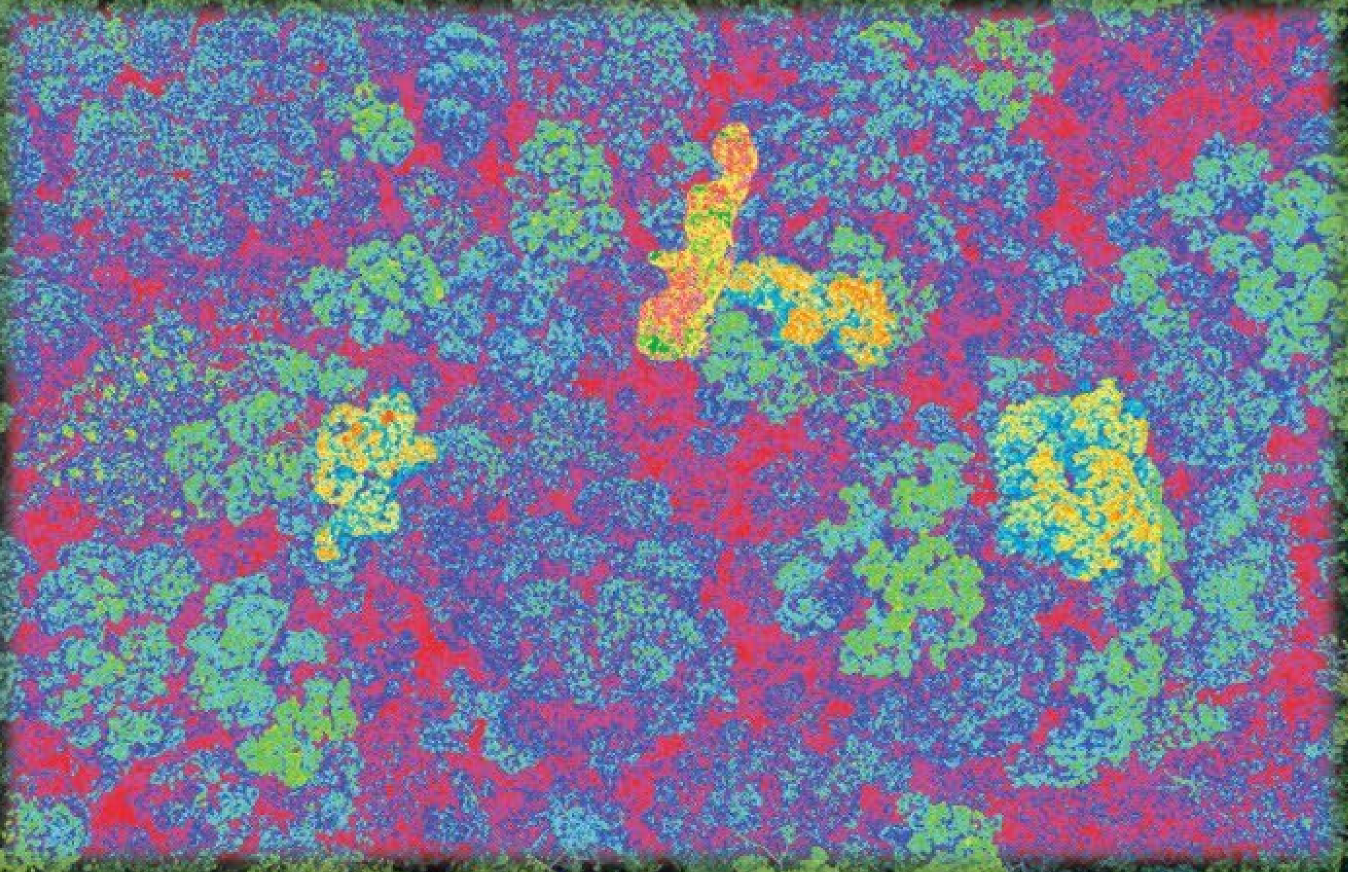


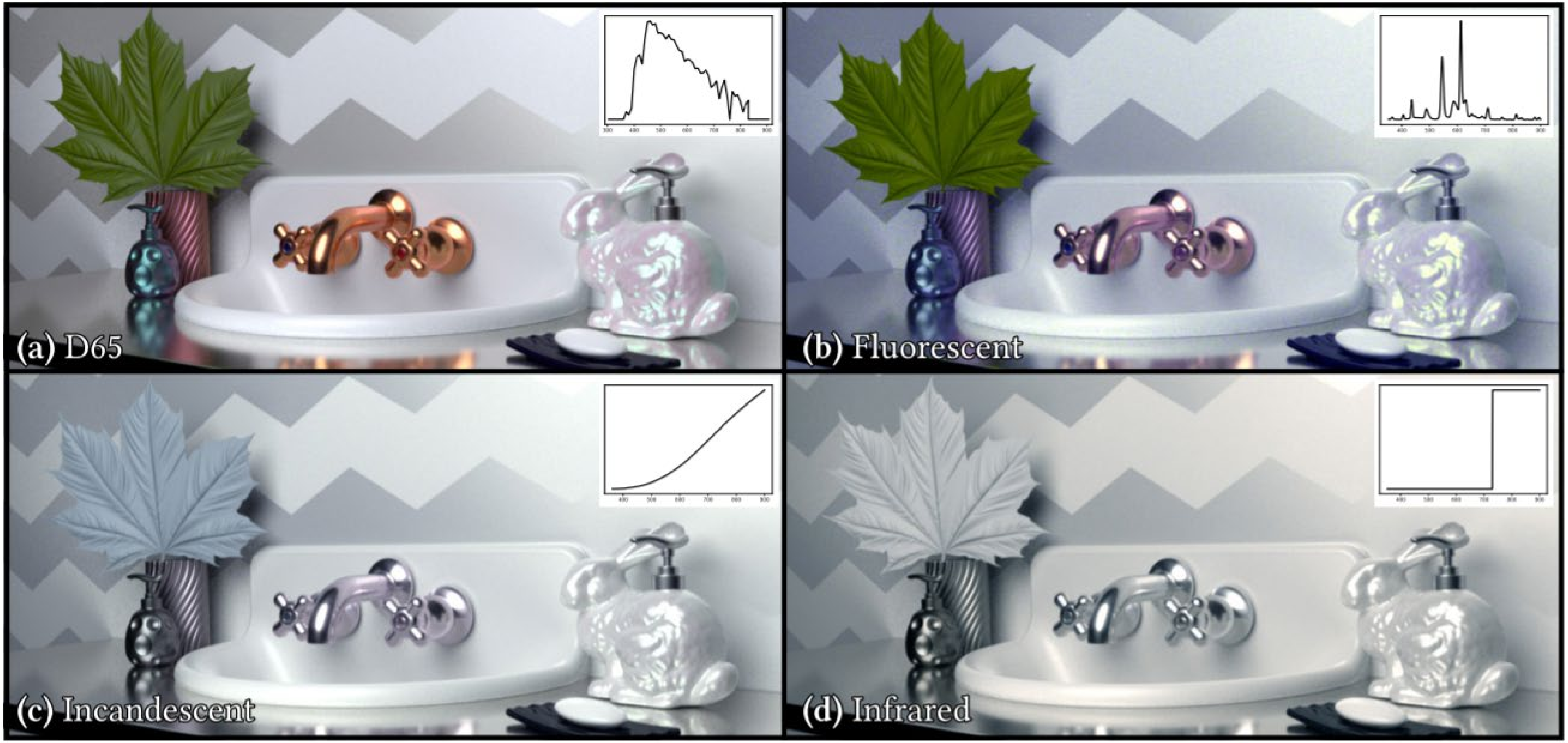


Source: Liu et al. 2017

Hyperspectral Imaging – *Where is it important?*

Remote Sensing





Source: [Dupuy and Jakob, 2018]

Source: [Alvarez-Cortes et al. 2015]

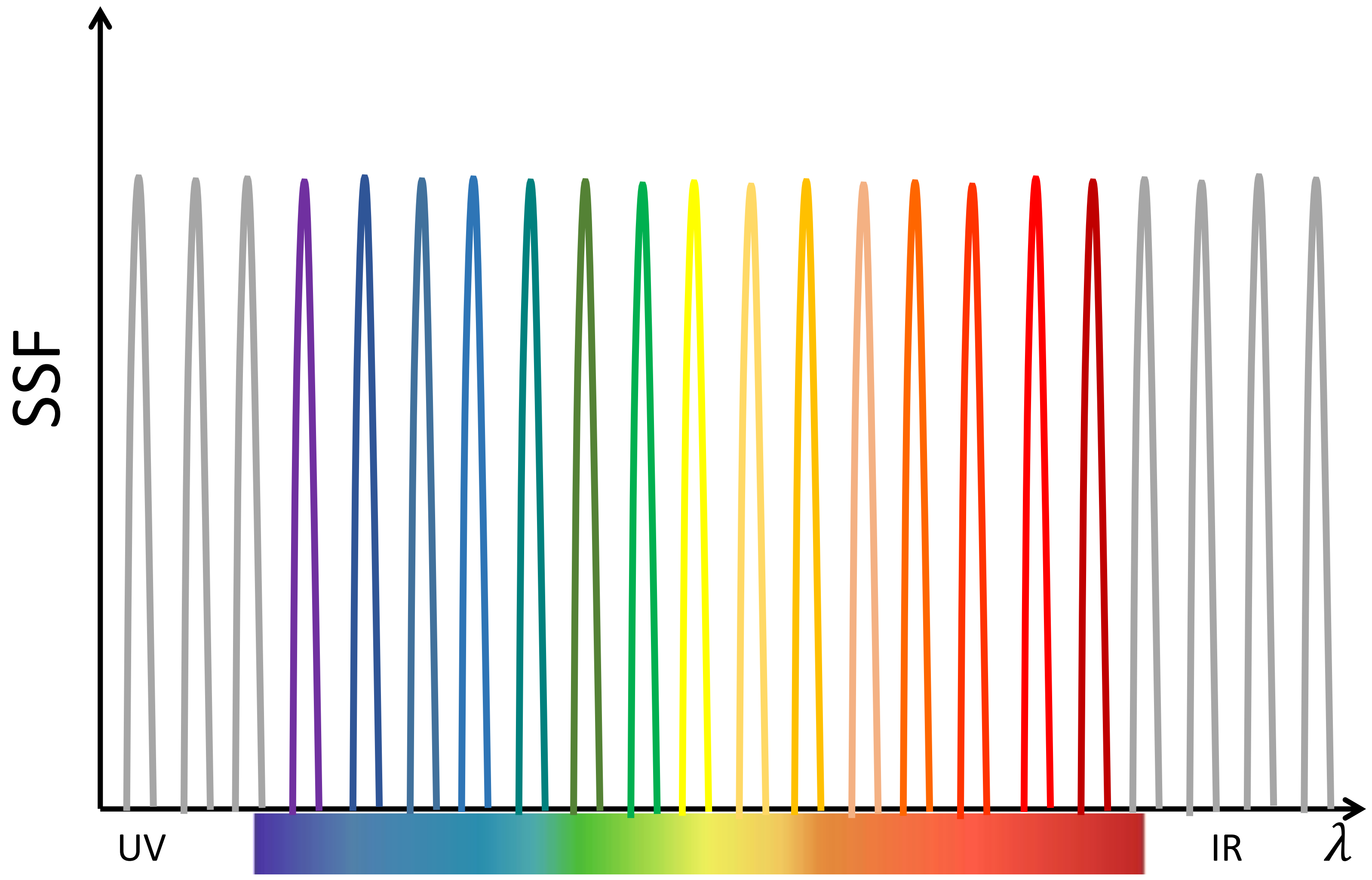
Dawn of the Planet of the Apes © 20th Century Fox

Questions?

Hyperspectral Imaging

- **What** is it?
- **Why** is it important? (a.k.a. applications)
- **How** do we capture it?
  - **Brainstorming!**

Hyperspectral Imaging – *The naïve approach*



Hyperspectral Imaging – *The naïve approach*

- **Note:** Hyperspectral cameras exists in the market
  - But they are very expensive (25K-100K)

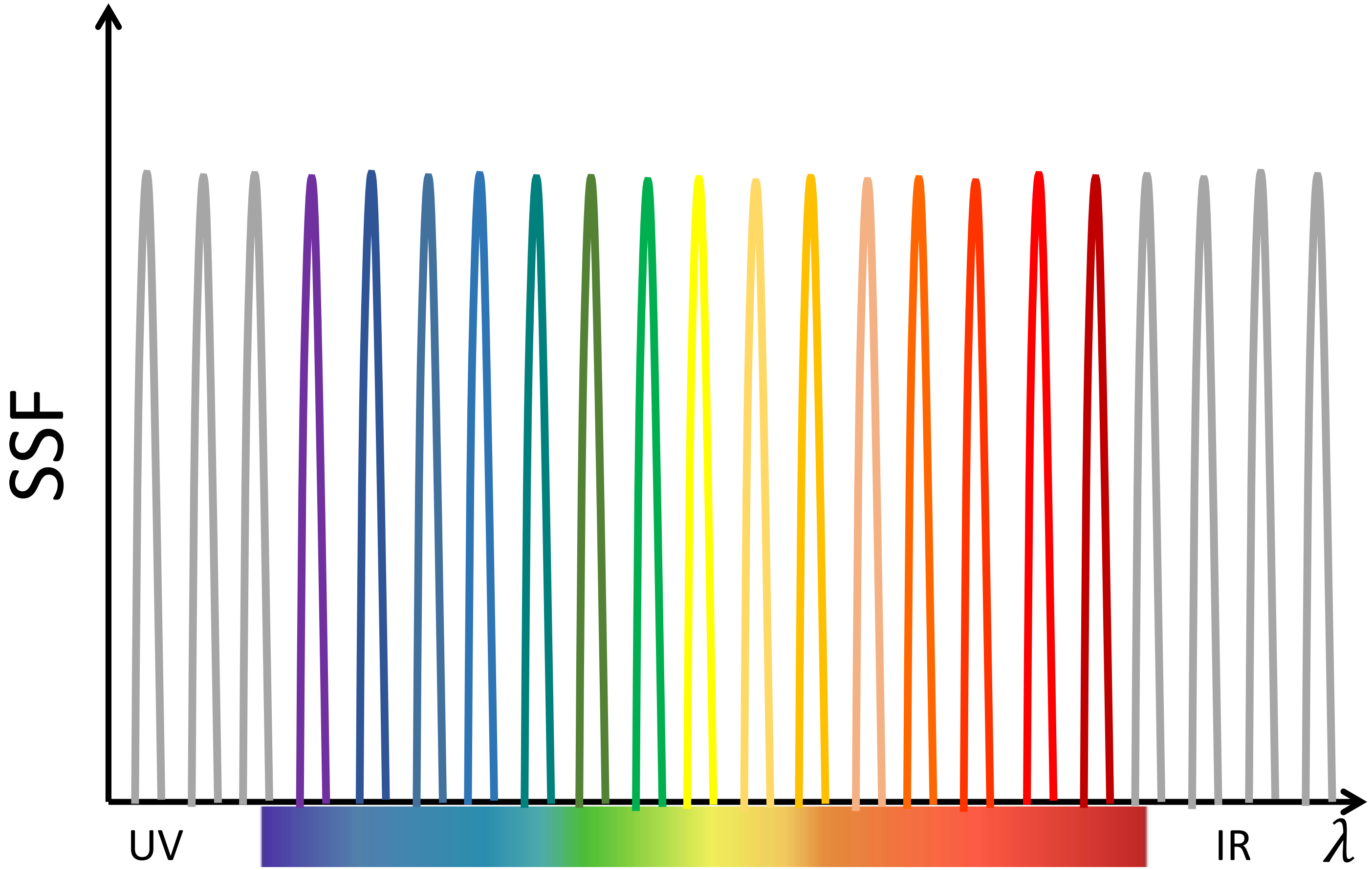
Bandpass filter
[Mansouri et al. 2007]

LCTF (liquid crystal tunable filter)
[Attas et al. 2003]

  - Static Scenes only
  - Low spatial resolution

Source: Choi et al. 2017

Hyperspectral Imaging – *The naïve approach*



- **Brainstorming**:

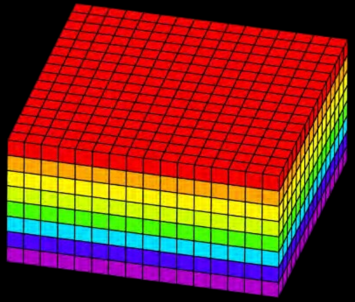
Can we do this with a conventional (i.e. no filters) camera?

Questions?

Hyperspectral Imaging - *How do we capture it?*

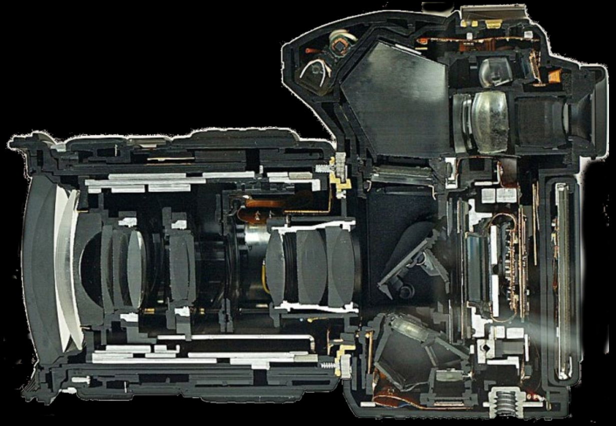
- **Spectral scanning** – *Naïve approach*
  - *Static scenes only, low spatial resolution*
- **Reconstruction-Based** – *The Comp. Img. Approach*
  - *Code the recorded signal; use computation for reconstruction*

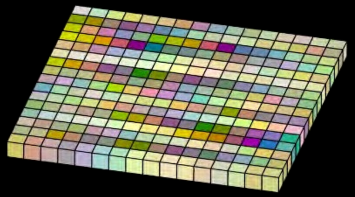
Reconstruction-Based Hyperspectral Imaging



Spectra

$\mathbf{h}$





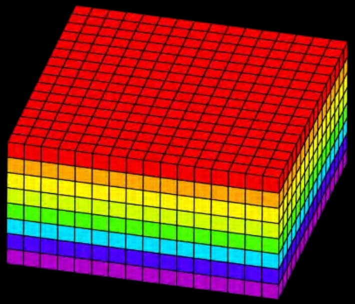
Observed image

$\mathbf{i}$

$$\underbrace{\mathbf{i}}_{\text{Observation (2D)}} = \underbrace{\Phi}_{\text{Light modulation (3D to 2D)}} \underbrace{\mathbf{h}}_{\text{Spectra (3D)}}$$

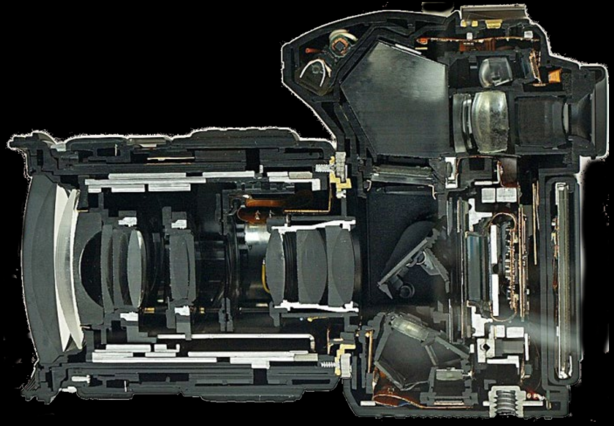
Source: Choi et al. 2017

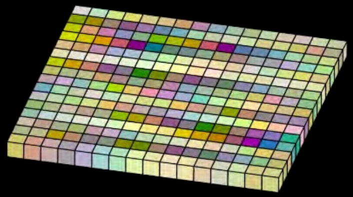
Reconstruction-Based Hyperspectral Imaging



Spectra

$\mathbf{h}$





Observed image

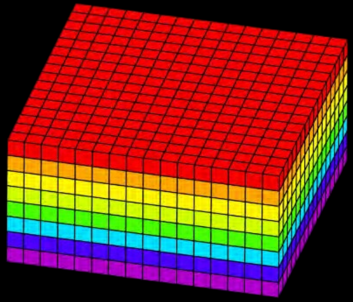
$\mathbf{i}$

$$\mathbf{i} = \Phi \mathbf{h}$$

Reconstruction is an inverse problem of optical imaging

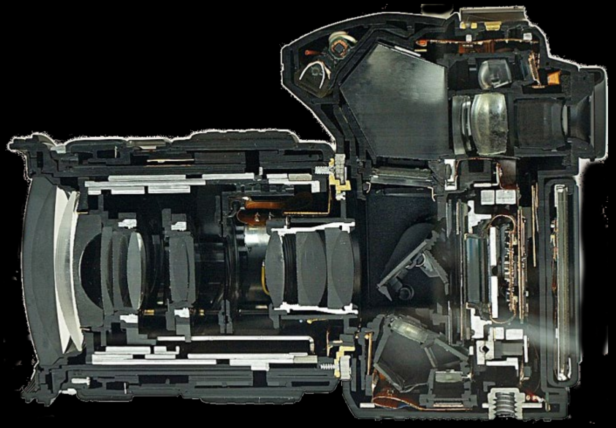
Source: Choi et al. 2017

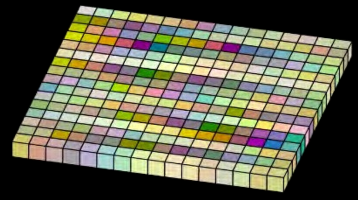
Reconstruction-Based Hyperspectral Imaging



Spectra

$\mathbf{h}$





Observed image

$\mathbf{i}$

$$\mathbf{h} = \Phi^{-1}\mathbf{i}$$

Reconstruction is an inverse problem of optical imaging

Source: Choi et al. 2017

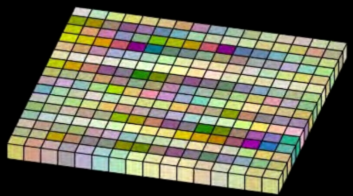
Reconstruction-Based Hyperspectral Imaging

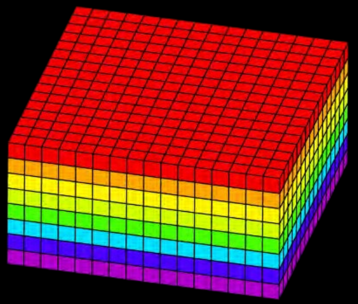
Key: Find a good image formation

$$\min_{\mathbf{h}} \left\| \mathbf{i} - \boldsymbol{\Phi}\mathbf{h} \right\|_2^2$$

"Find a hyperspectral image $\mathbf{h}$ that satisfies the image formation"

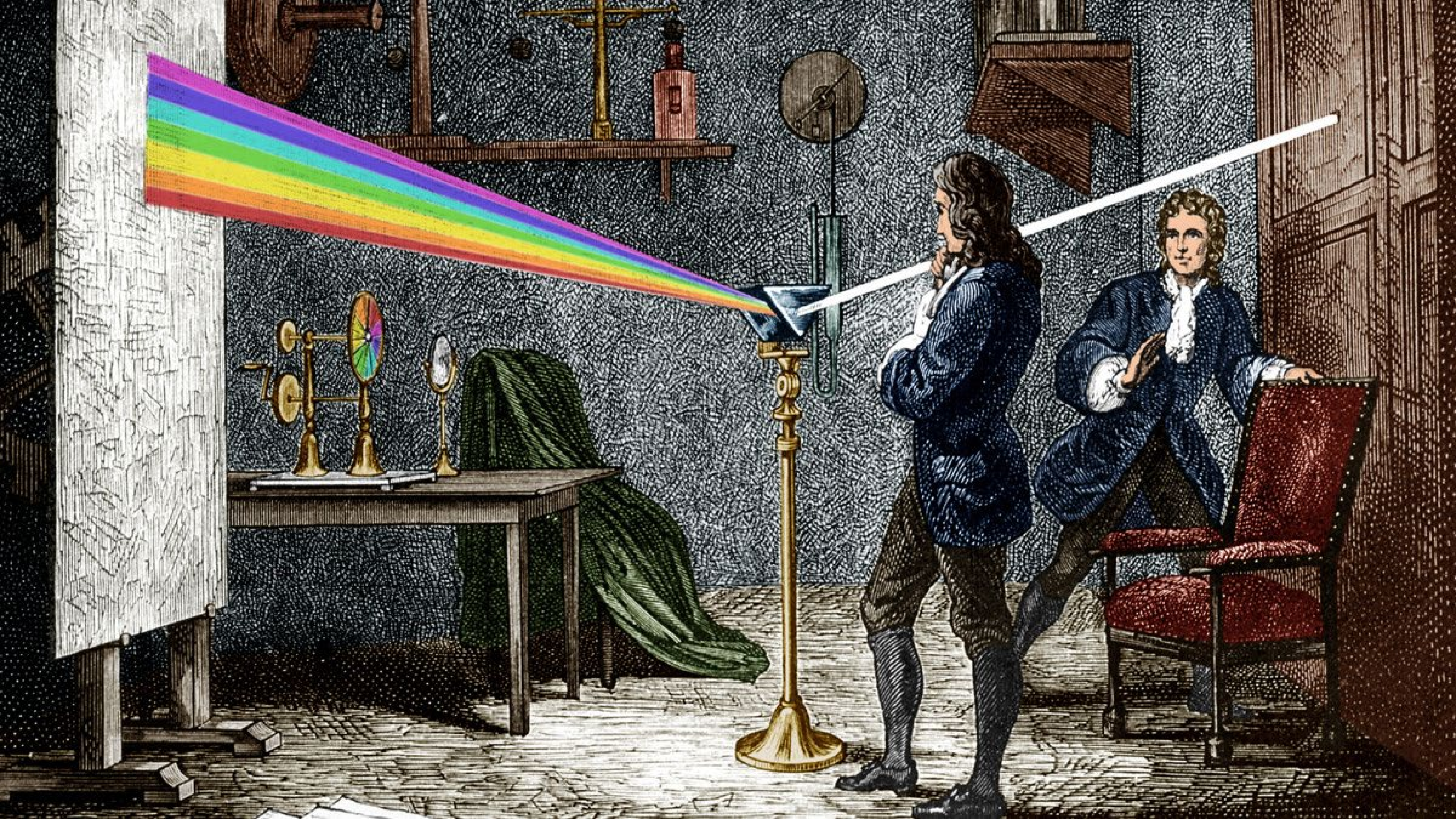
$\#$ equations $\ll$ $\#$ unknowns





underdetermined system

Source: Choi et al. 2017



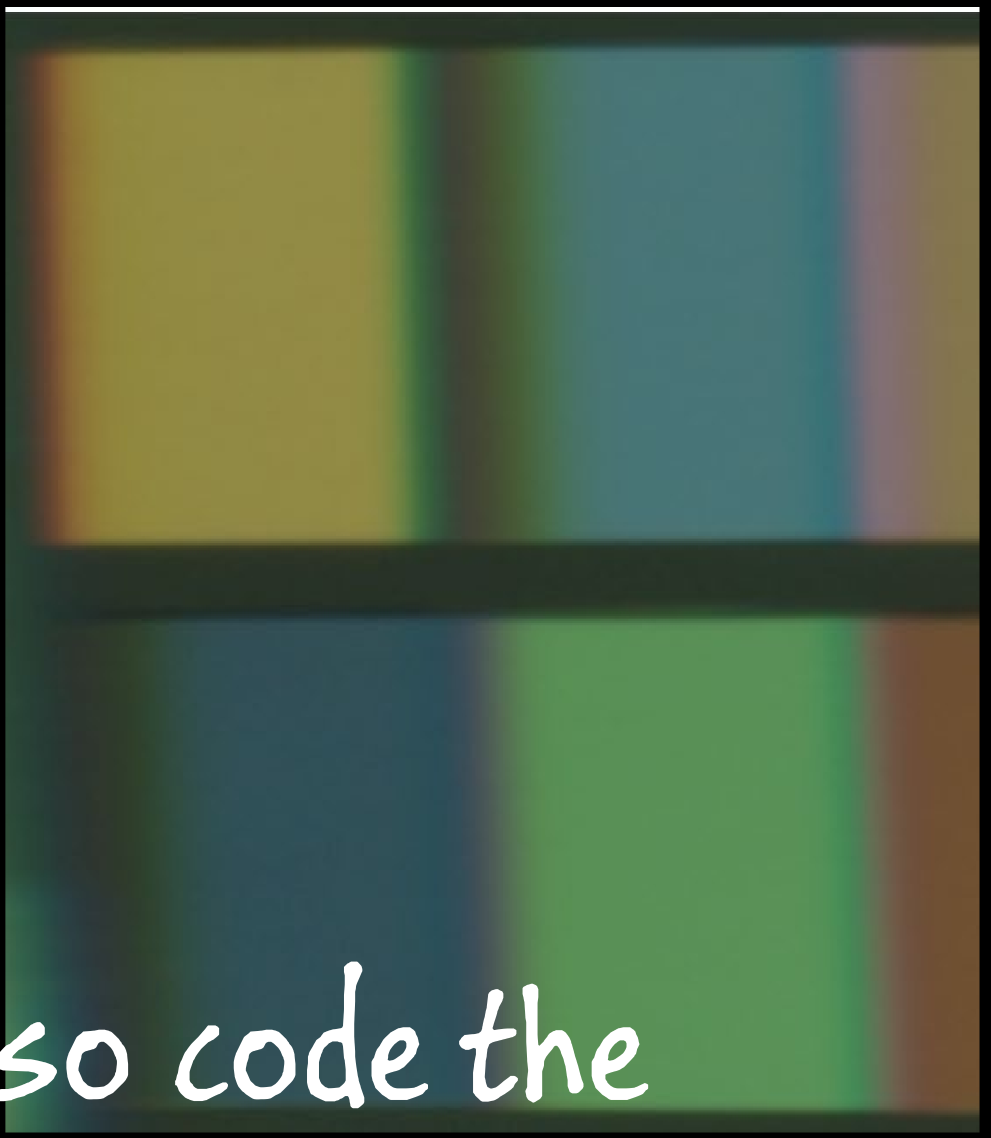
Reconstruction-Based Hyperspectral Imaging

Scene spectra

Shear

Projection

Source: Choi et al. 2017

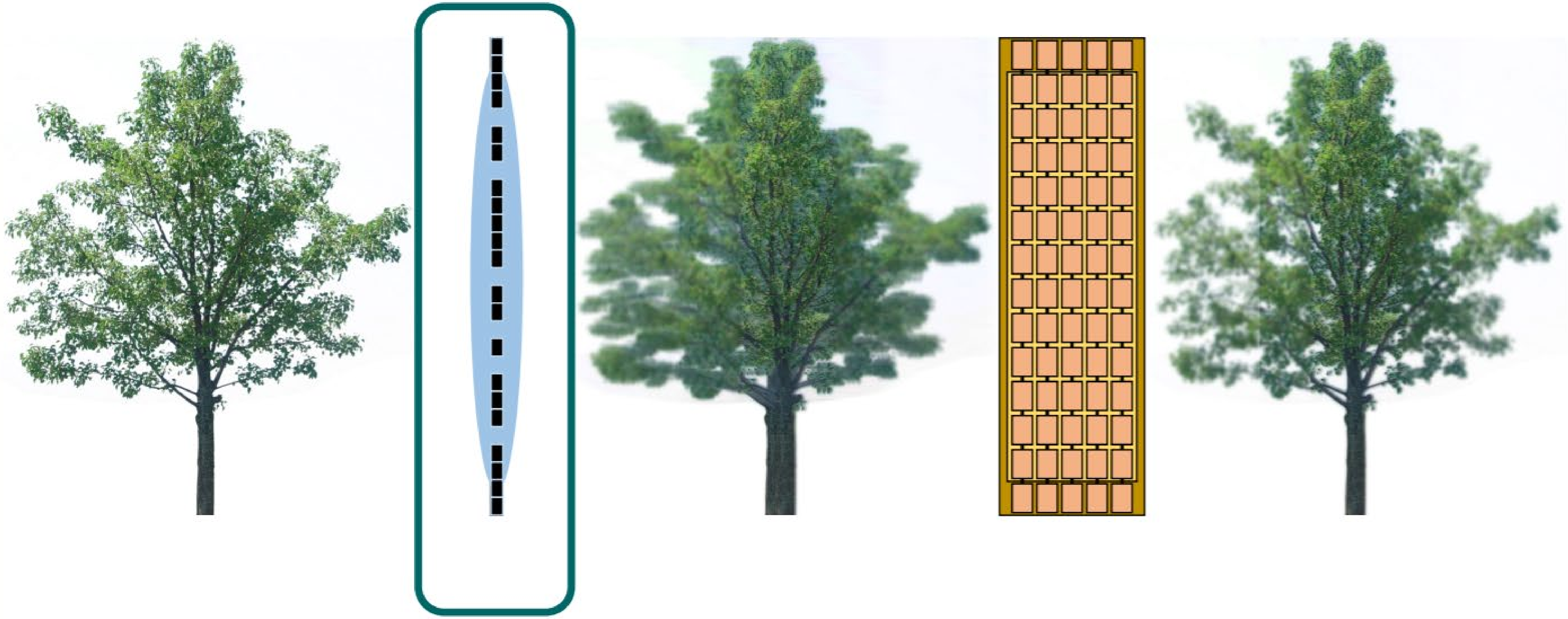


But... what if we also code the aperture?

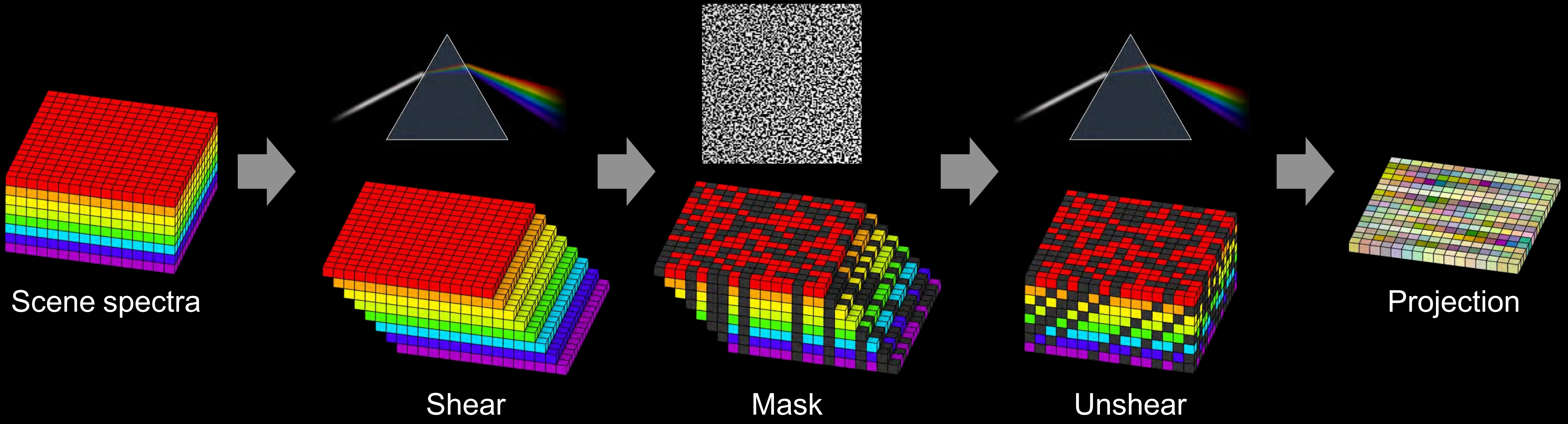
Source: Baek et al. 2017

Remember...

Last class: Coded apertures for defocus deblurring

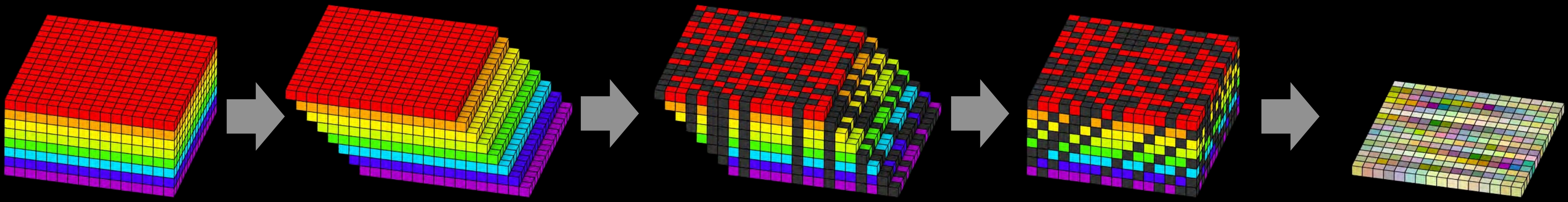


Reconstruction-Based Hyperspectral Imaging



Source: Choi et al. 2017

Reconstruction-Based Hyperspectral Imaging



Spectra $\mathbf{h}$

Shear

Mask

Unshear and project

Observed image $\mathbf{i}$

$$\mathbf{i} = \Phi \mathbf{h}$$

Observation (2D)

Light modulation (3D to 2D)

Spectra (3D)

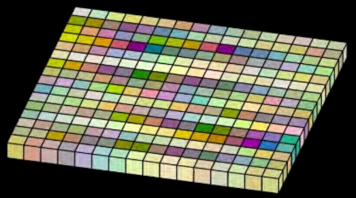
Source: Choi et al. 2017

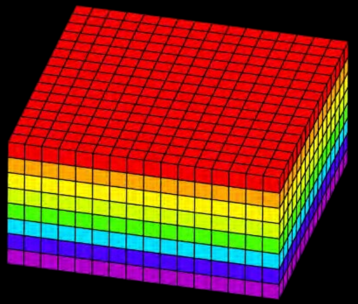
Reconstruction-Based Hyperspectral Imaging

$$\min_{\mathbf{h}} \left\| \mathbf{i} - \mathbf{\Phi} \mathbf{h} \right\|_2^2$$

“Find a hyperspectral image $\mathbf{h}$ that satisfies the image formation”

\# equations $\ll$ # unknowns





underdetermined system

Source: Choi et al. 2017

Optimization-Based Hyperspectral Imaging

- TV-L1 is very common in computational photography

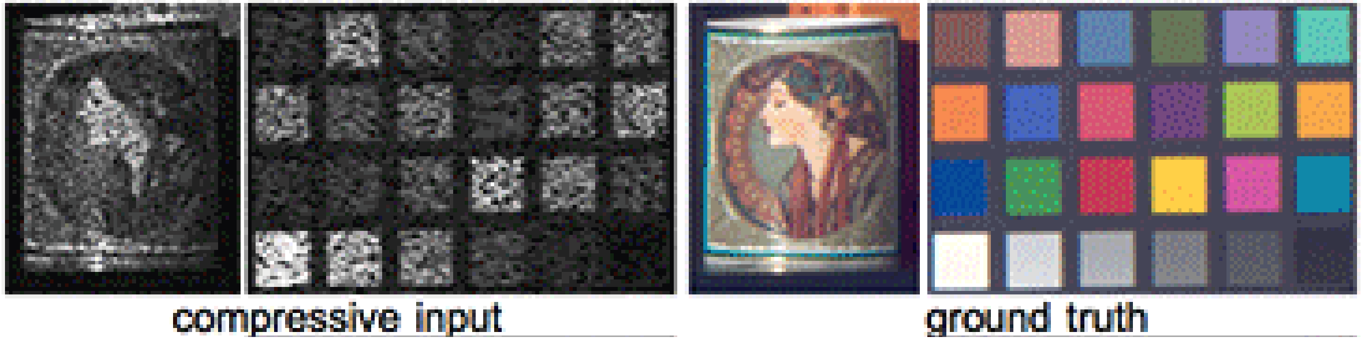
$$\min_{\mathbf{h}} \left\| \mathbf{i} - \Phi \mathbf{h} \right\|_2^2 + \left\| \qquad \mathbf{h} \right\|_1$$

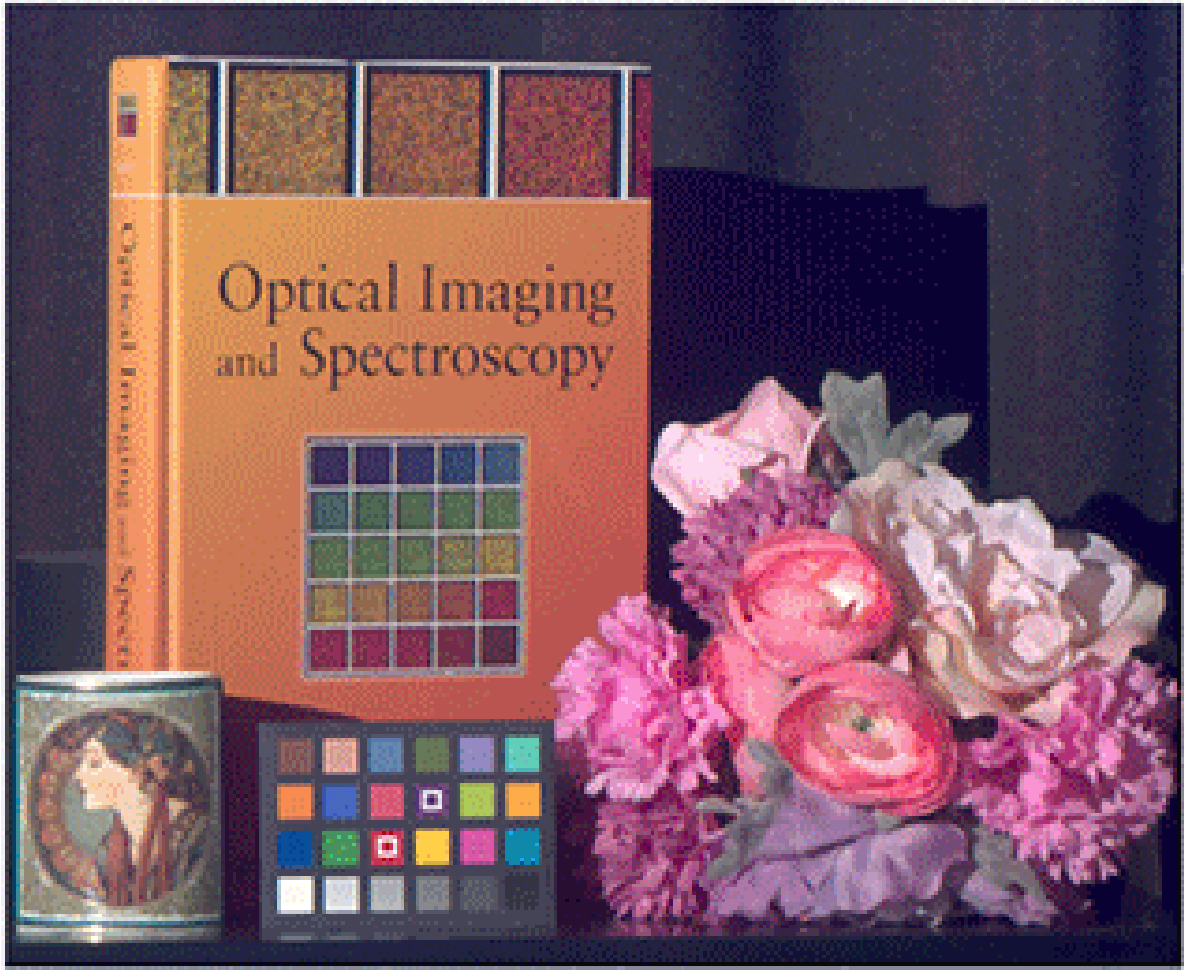
TwIST [Bioucas-Dias and Figueiredo 2007]

SpaRSA [Wright et al. 2009]

Source: Choi et al. 2017

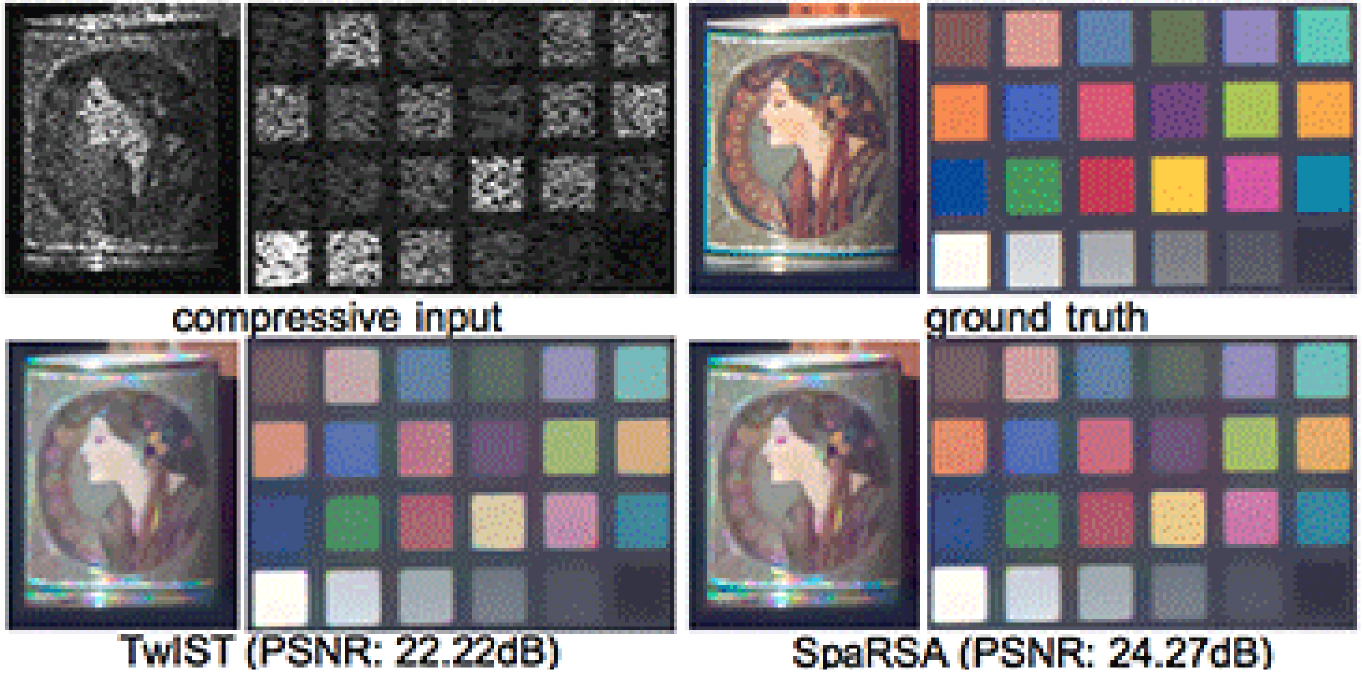
Reconstruction-Based Hyperspectral Imaging

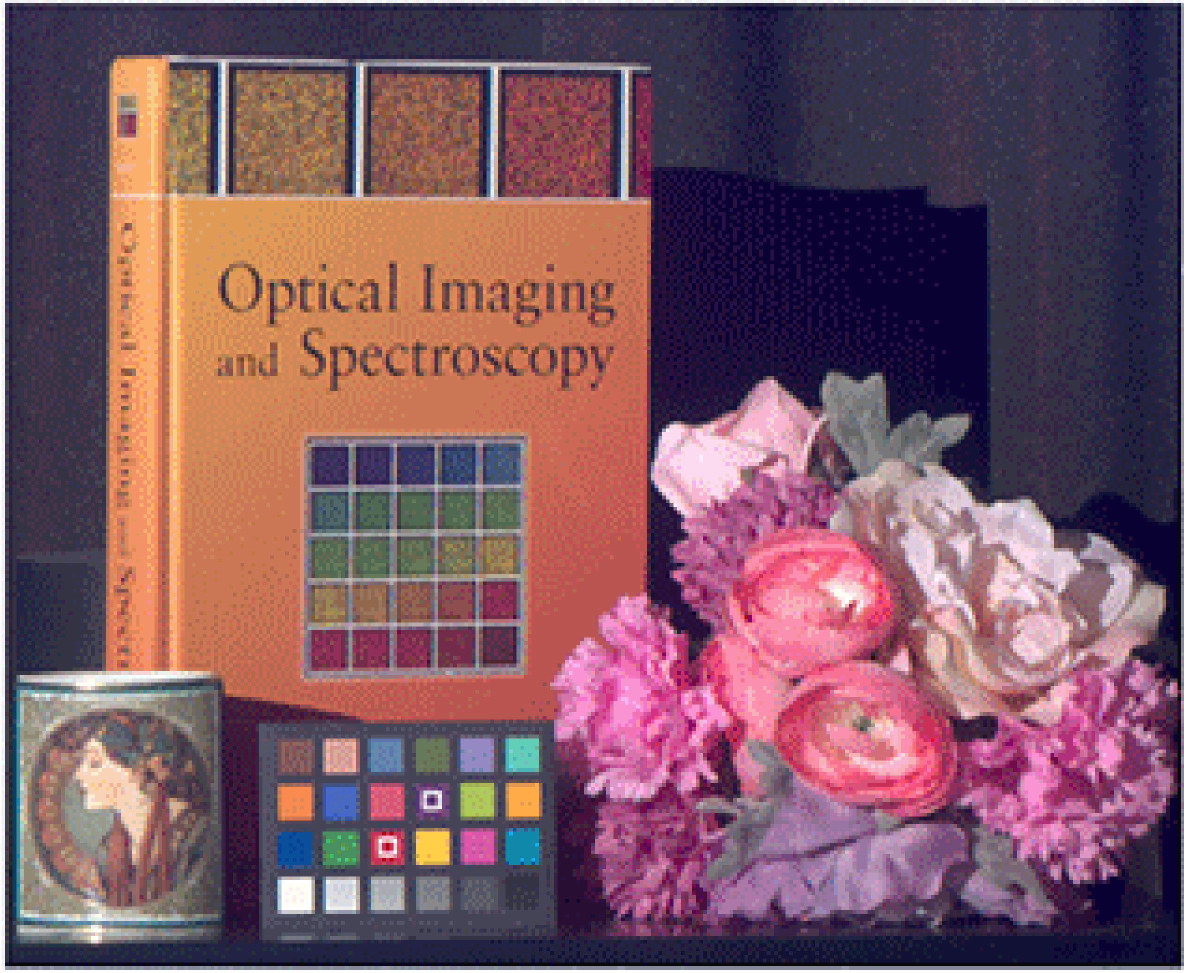




Choi et al. 2017

Reconstruction-Based Hyperspectral Imaging



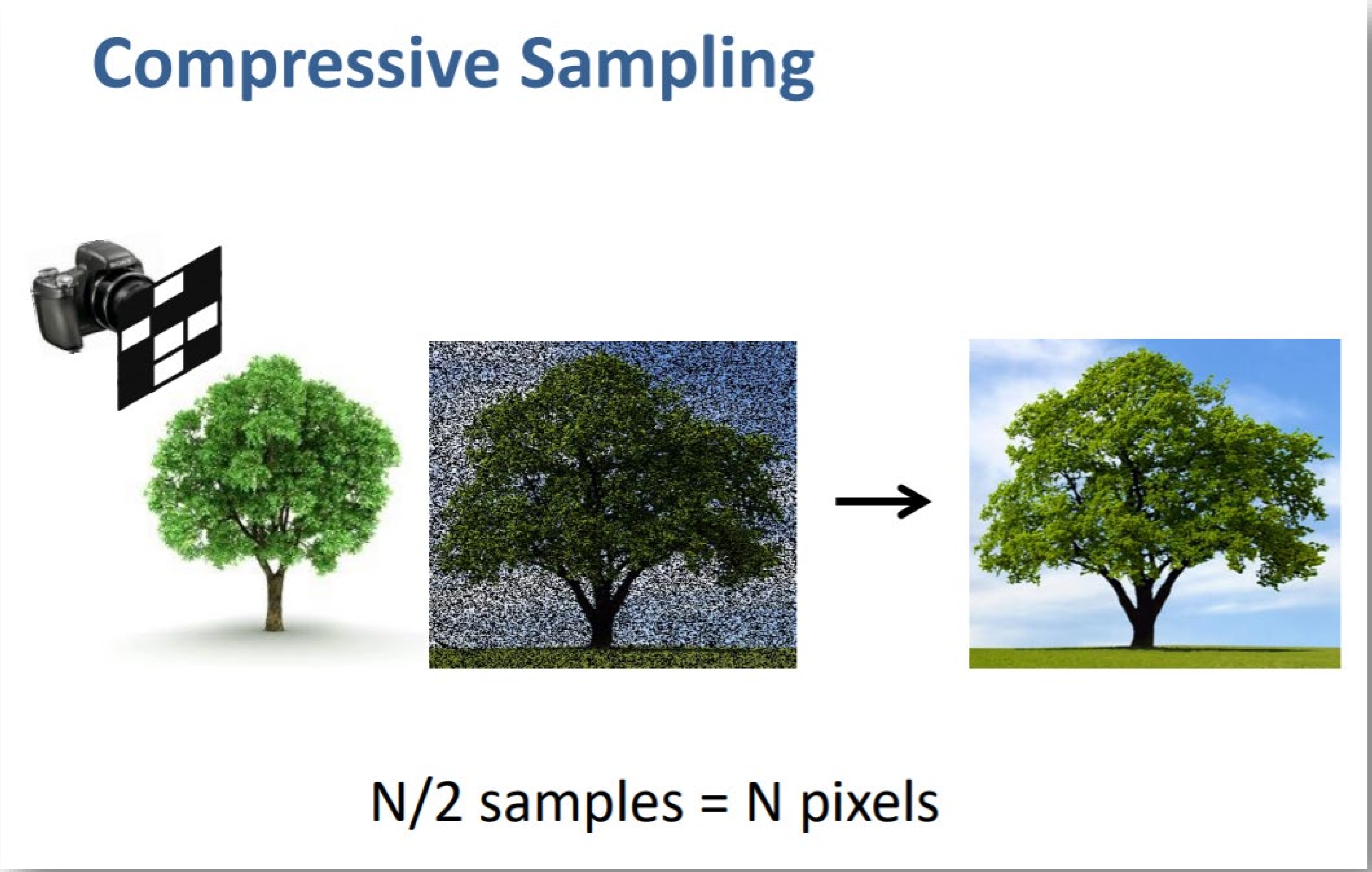


Can we do better?

Choi et al. 2017

Can we do better than Total Variation?

Remember...



Compressed Hyperspectral Imaging

- Use an overcomplete dictionary and a sparse code to represent a data

$$\mathbf{h} = \mathbf{D}\boldsymbol{\alpha}$$

D: a dictionary

α: a sparse code

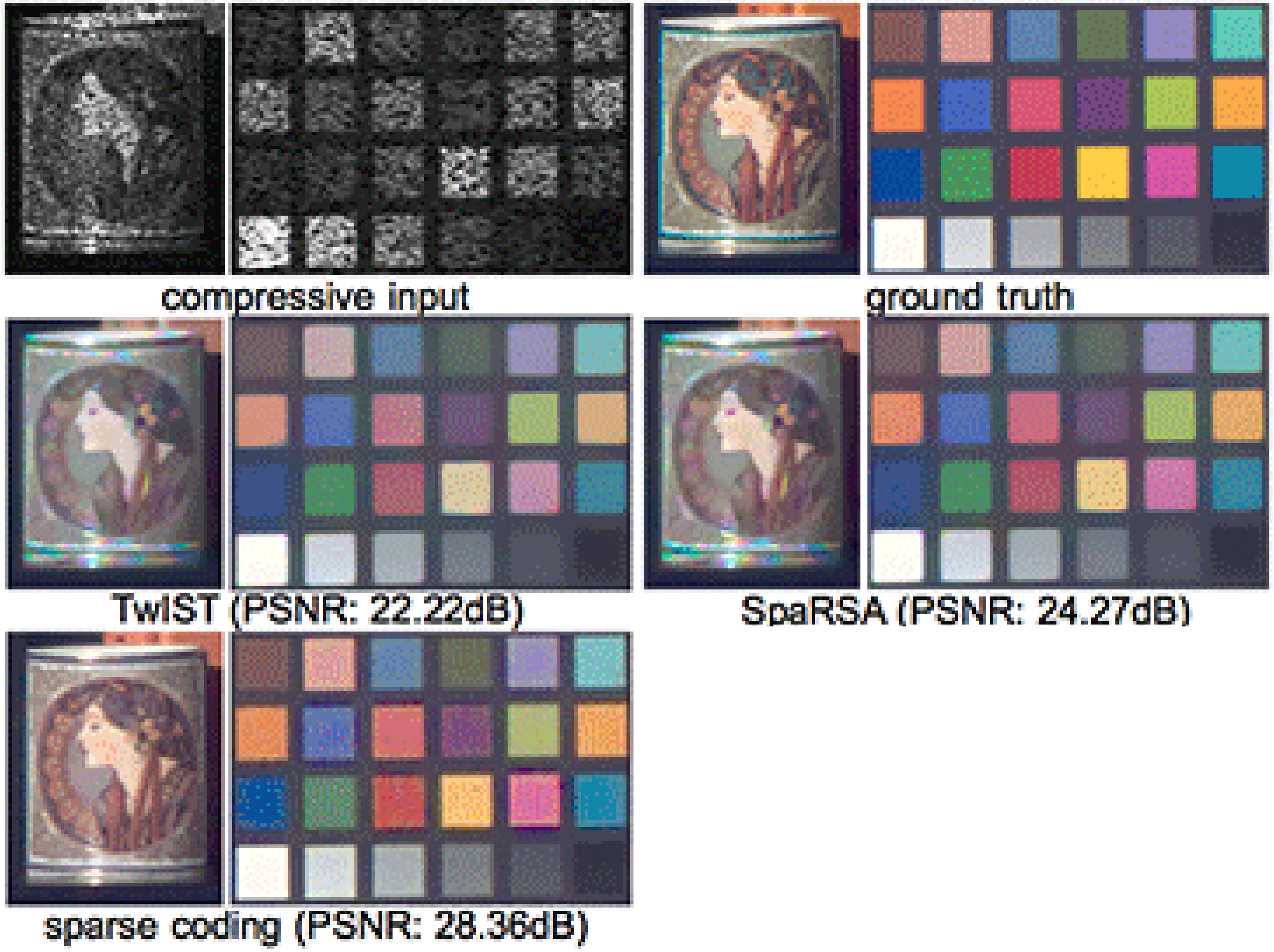
For all overlapping image patches

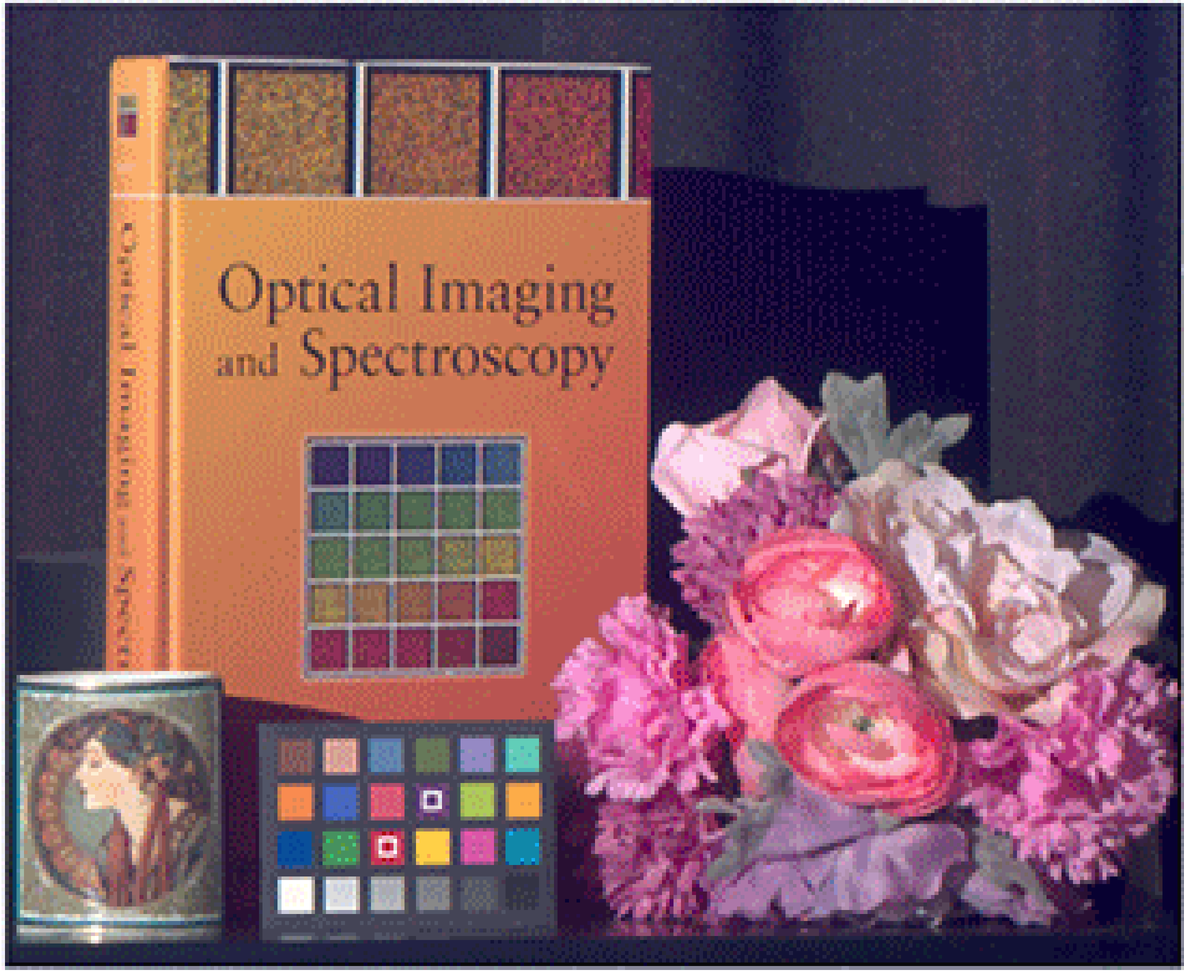
$$\min_{\alpha} \left\| \mathbf{i} - \boldsymbol{\phi}\mathbf{D}\boldsymbol{\alpha} \right\|_2^2 + \left\| \boldsymbol{\alpha} \right\|_1$$

[Lin et al. 2014]

Source: Choi et al. 2017

Reconstruction-Based Hyperspectral Imaging





Choi et al. 2017

Hyperspectral Imaging - *How do we capture it?*

- **Spectral scanning** – *Naïve approach*
  - *Static scenes only, low spatial resolution*
- **Reconstruction-Based** – *The Comp. Img. Approach*
  - *TV-based optimization, CS, Deep Learning…*
  - *Bulky hardware, still expensive*

Questions?

Hyperspectral Imaging - *How do we capture it?*

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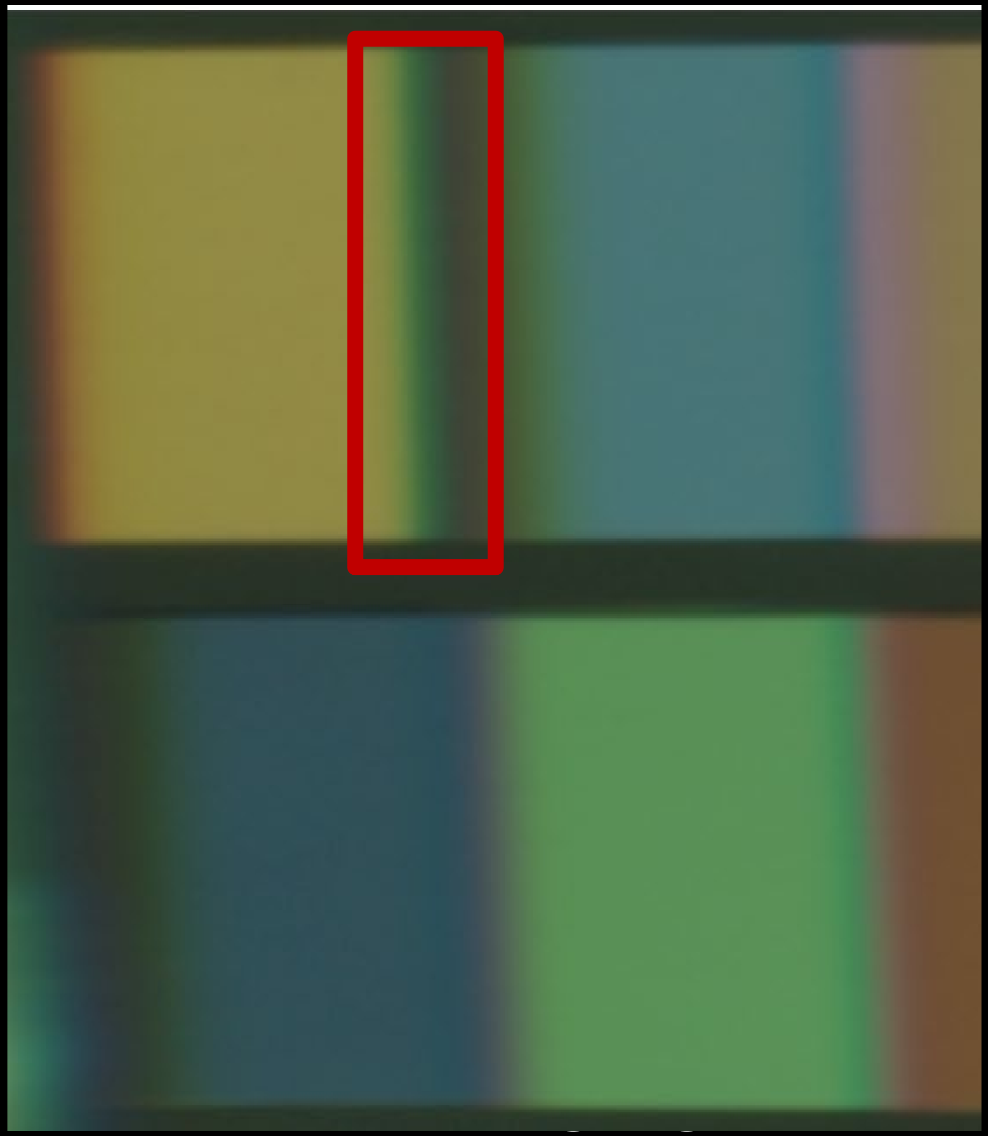
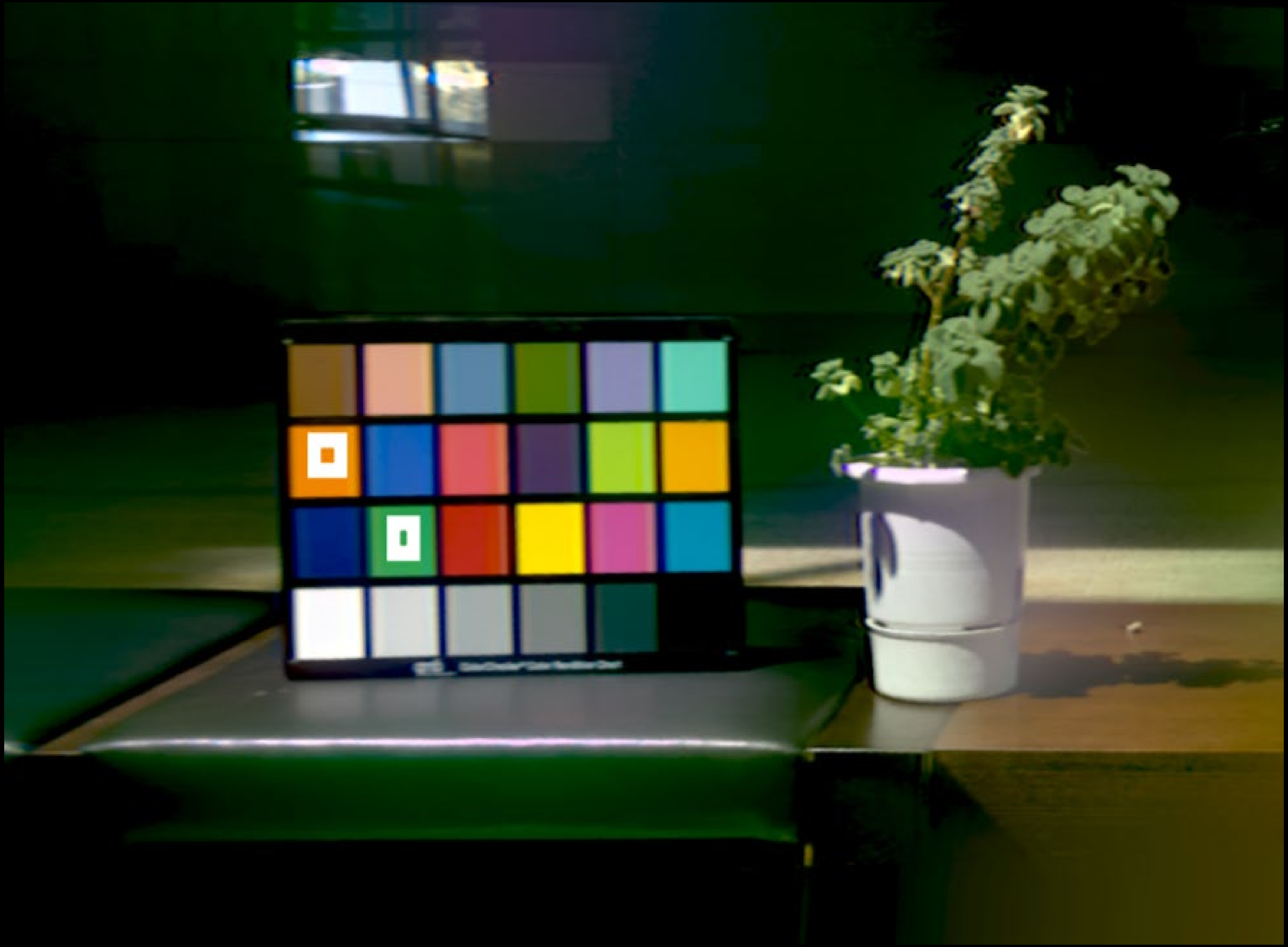


Source: Baek et al. 2017



Source: Baek et al. 2017

Hyperspectral Imaging - *How do we capture it?*

- **Spectral scanning** – *Naïve approach*
  - *Static scenes only, low spatial resolution*
- **Reconstruction-Based** – *The Comp. Img. Approach*
  - *TV-based optimization, CS, Deep Learning…*
  - *Bulky hardware, still expensive*
- **High-Rez Dispersion-Based Imaging**
  - *Exploiting sparsity on the gradient domain!*
  - *See [Baek et al. 2017]*

Recap

- **Hyperspectral imaging**
  - **What:** Capturing more spectral samples, potentially outside the visible spectrum, than convential cameras/the human eye.
  - **Why:** Many applications in CV and Graphics, including material recognition, remote sensing, relighting…
  - **How:** From naïve spectral scanning to compressed sensing & deep learning.

- **What comes next?**

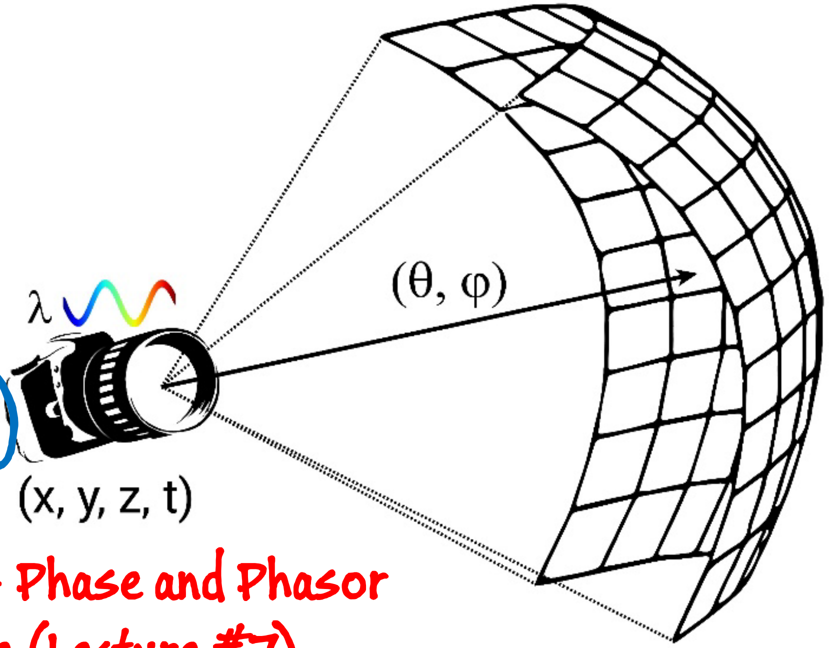
Questions?

James Clerk Maxwell (1831-1979)

Extended Plenoptic imaging

Hyperspectral imaging (Lecture #8)

$$P = P(\theta, \phi, \lambda, t, x, y, z, \varphi, \{s, p\})$$



Polarization - Phase and Phasor Amplitude (Lecture #7)

- Why are these interesting/important?
- How can we capture them back with our camera.

POLARIZATION IMAGING

OR ITS FANCIER NAME:

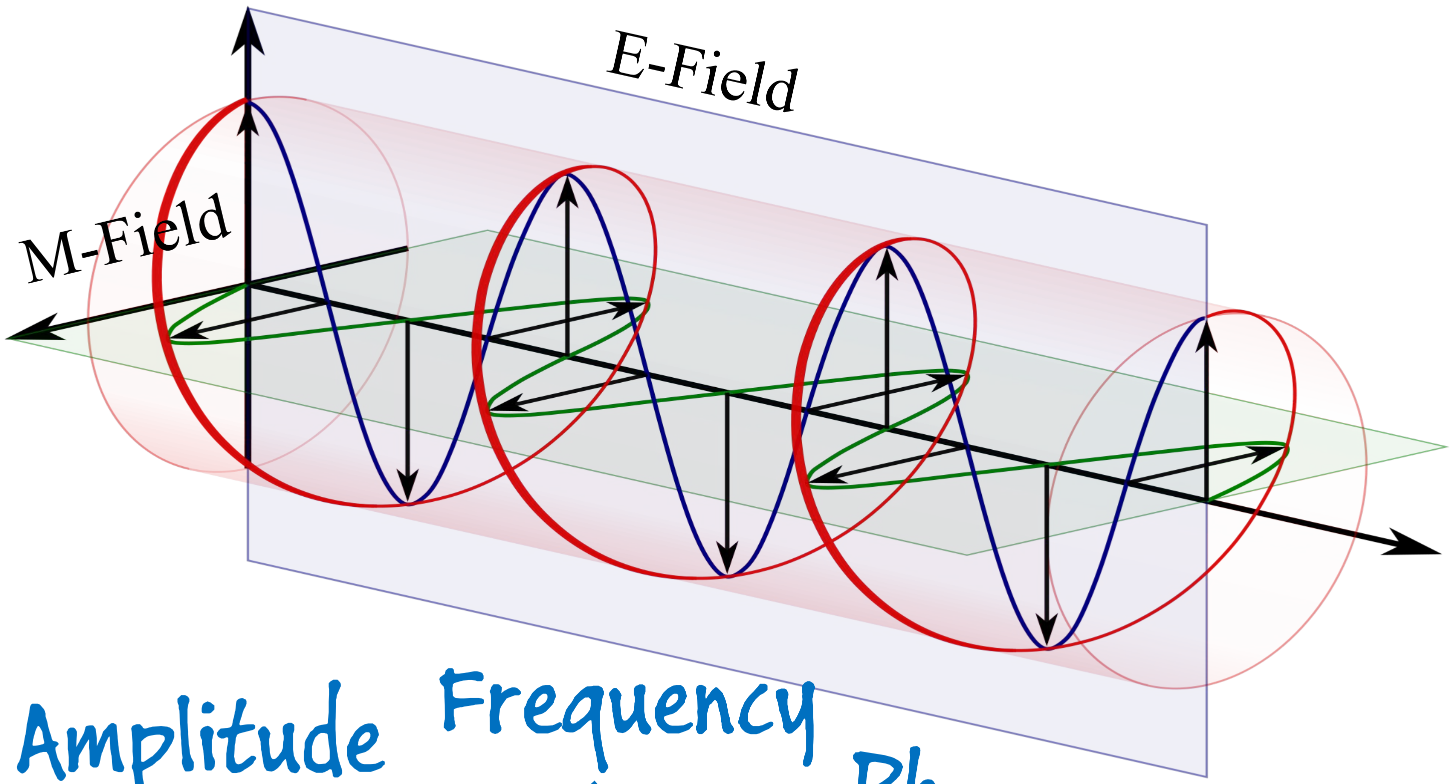
“STOKES IMAGING”

Source: Fredo Durand

Polarization Imaging

- **What** is it?
- **Why** is it important? (a.k.a. applications)
- **How** do we capture it?

Light as an electromagnetic wave

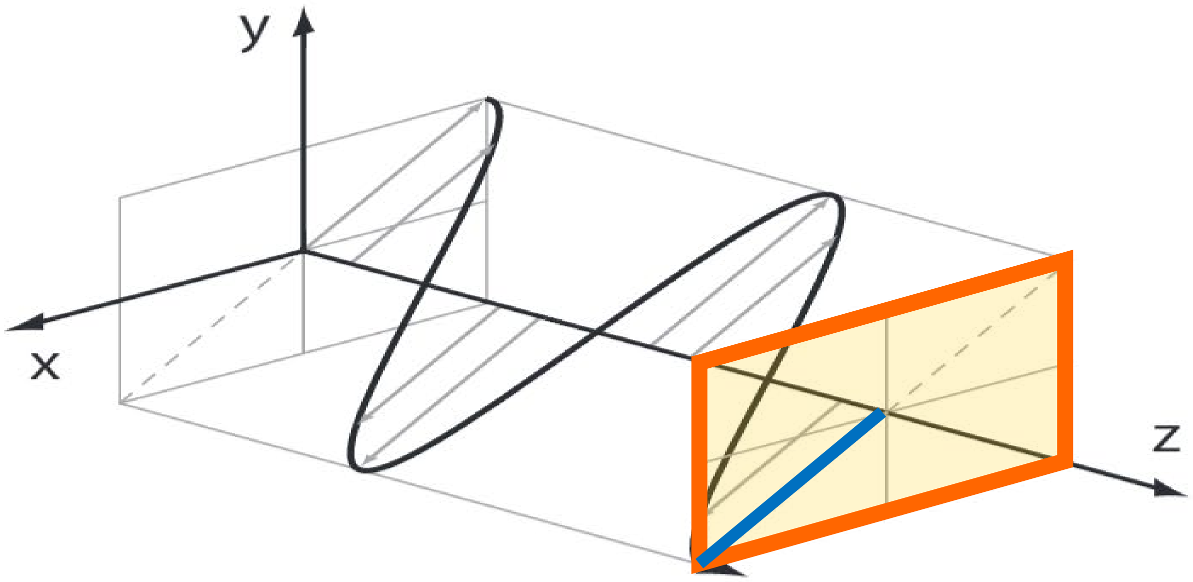


Amplitude Frequency Phase

$$E_{\mathbf{xy}}(r) = \langle I_s \sin(\omega r + \phi_s), I_p \sin(\omega r + \phi_p) \rangle$$

Polarization

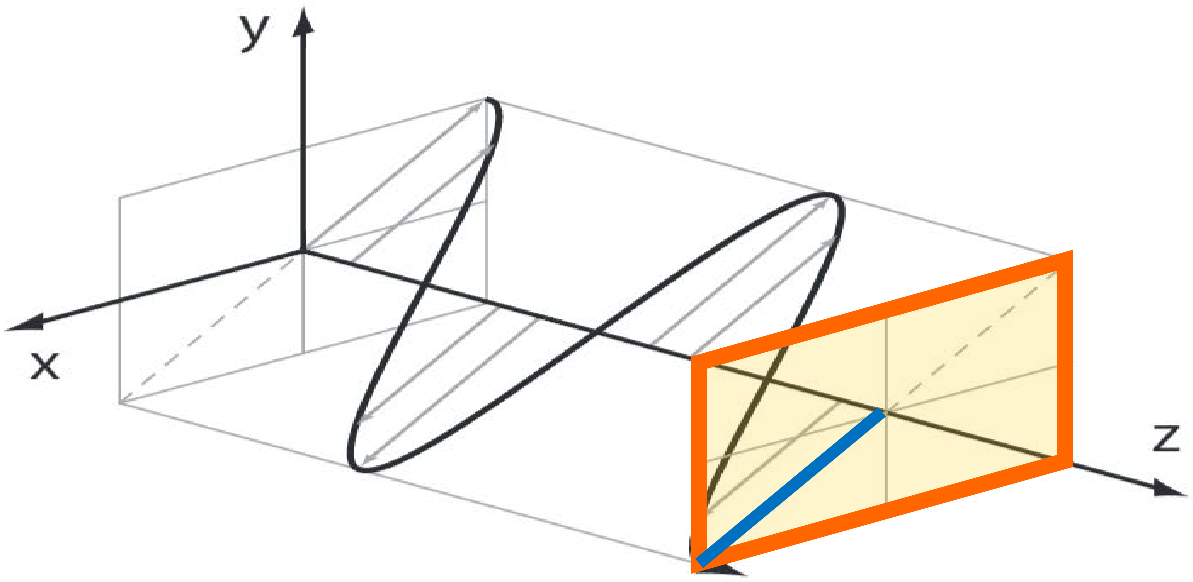
- The amplitude of both phasors and their relative phase, affect the *shape* of the wavefront – its polarization.



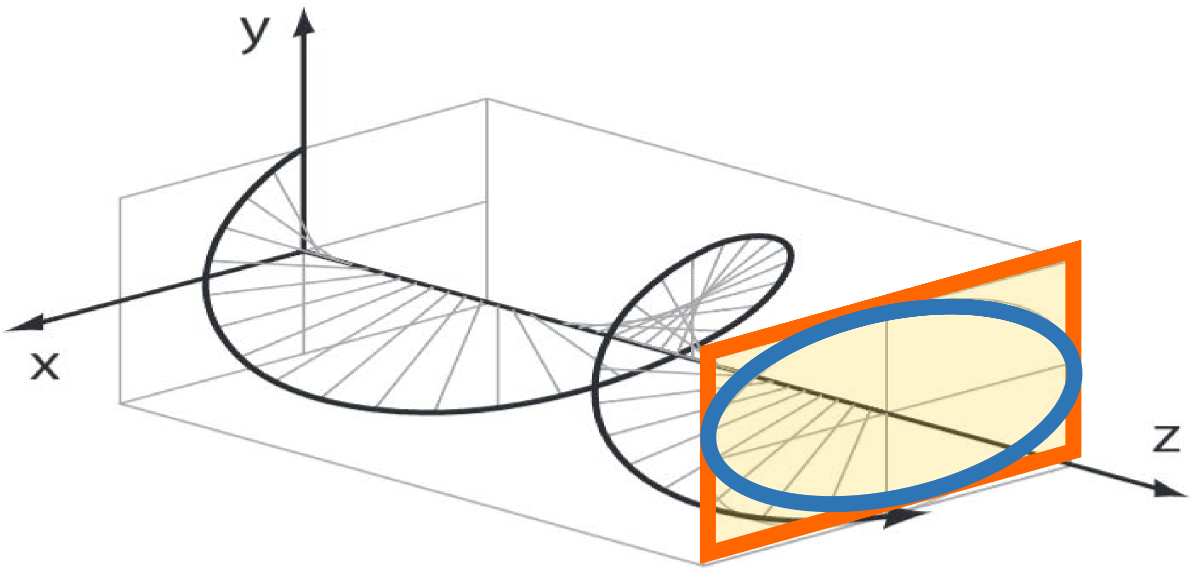
Linear polarization

Polarization

- The amplitude of both phasors and their relative phase, affect the *shape* of the wavefront – its polarization.

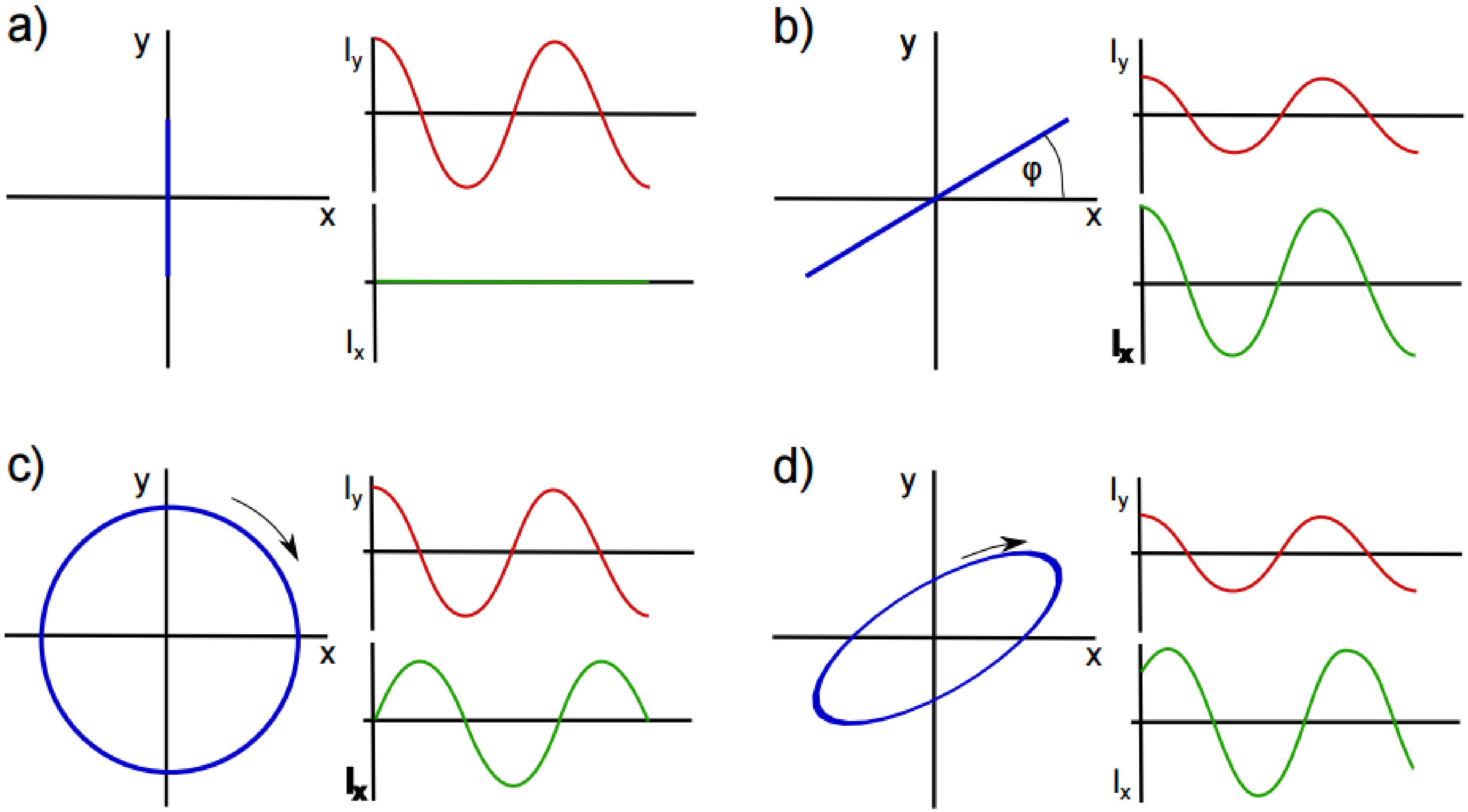


Linear polarization



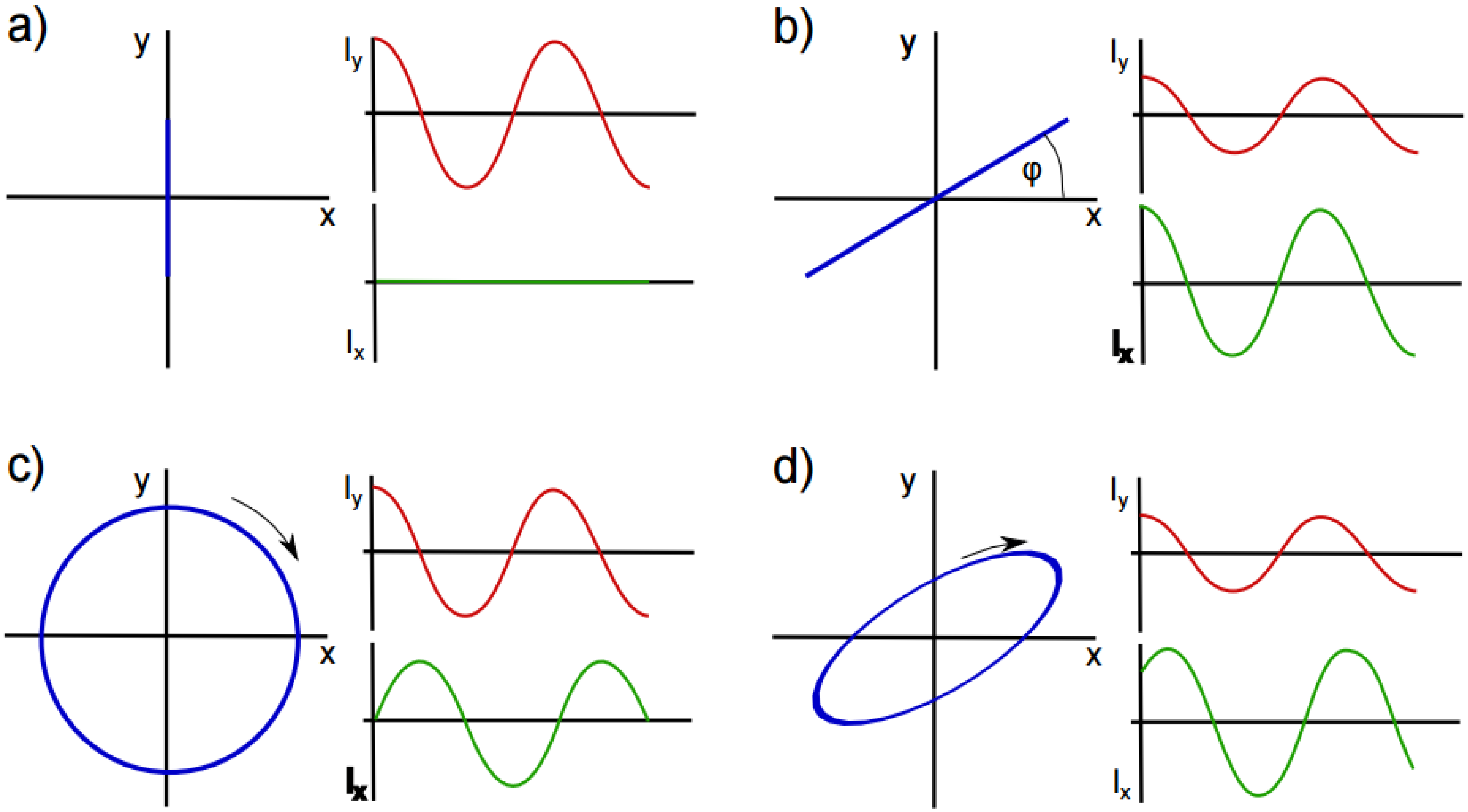
Elliptical polarization

Polarization – *A small quiz*



Polarization – *A small quiz*

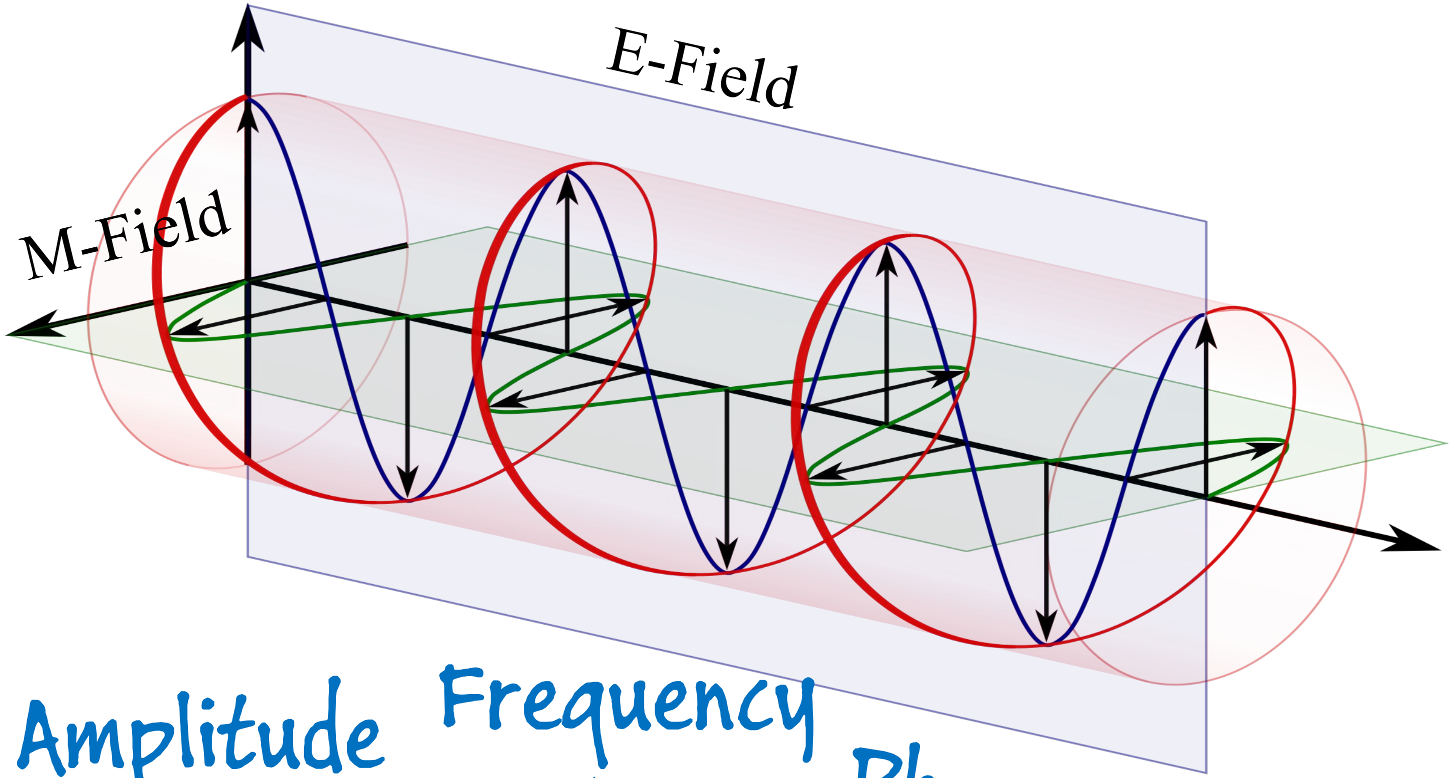
Linear polarization – In phase



Elliptical polarization – Off-phase

If $\mathrm{I_s} = \mathrm{I_p}$ and $\phi_s - \phi_p = 90º$ then circular polarization

Light as an electromagnetic wave



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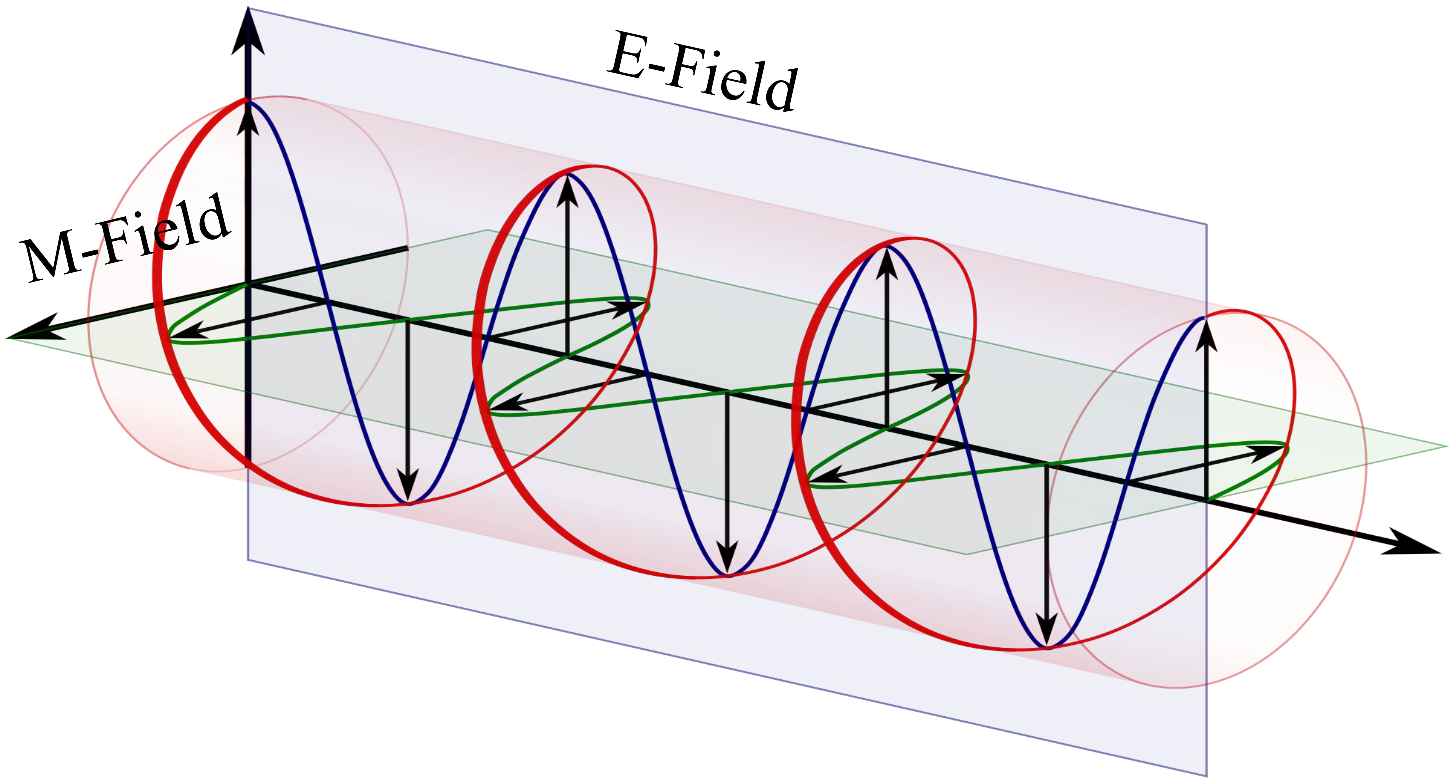
Light as an electromagnetic wave

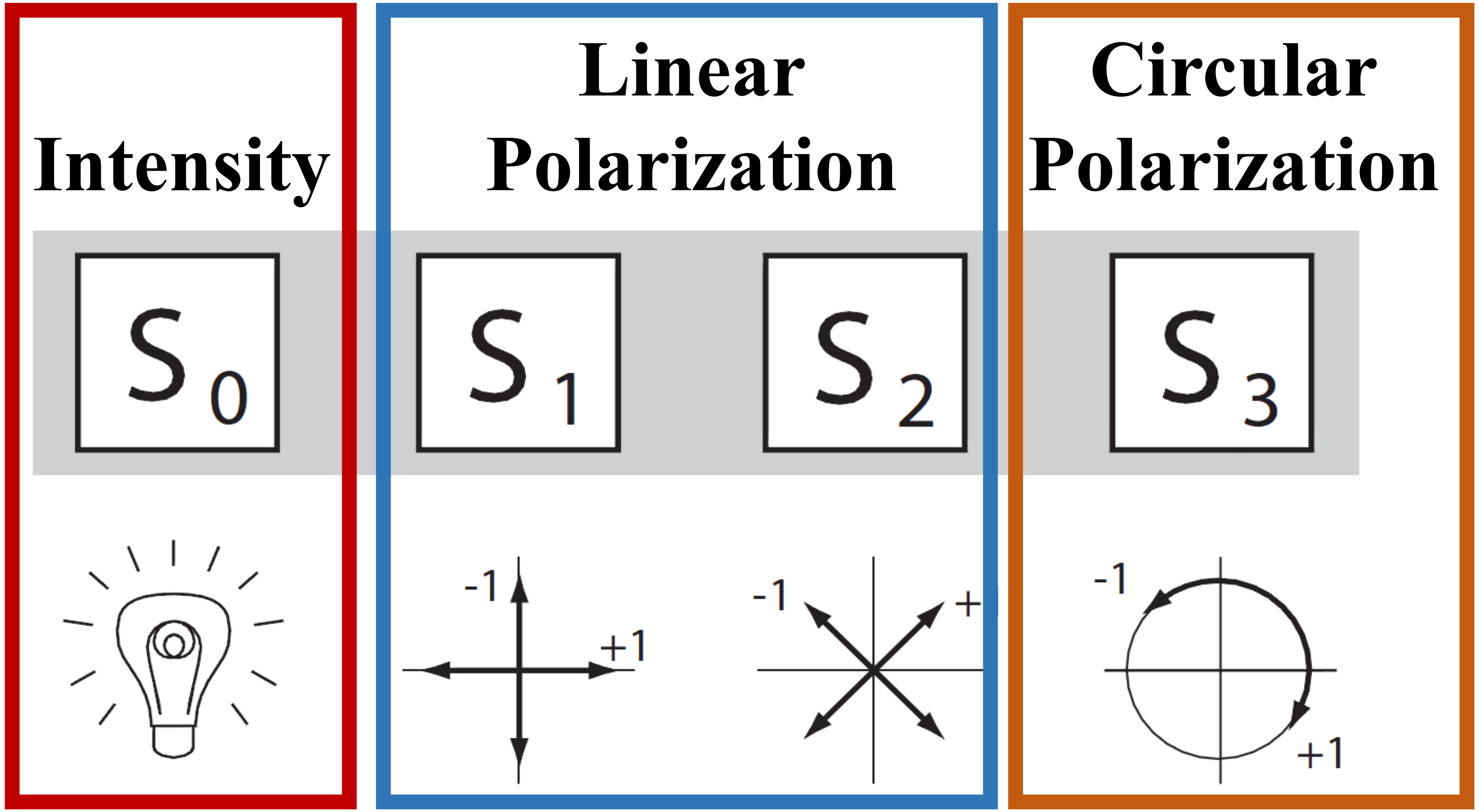


But wait, natural light is not completely polarized

We need another form of representing pol/unpolarized light!

Stokes Vectors

Sir George Stokes (1819-1903)



G. G. Stokes, (1852). "*On the composition and resolution of streams of polarized light from different sources*". Transactions of the Cambridge Philosophical Society

Stokes Vectors

Sir George Stokes (1819-1903)

Representing E as a complex number **(a.k.a. Jones vector):**

$$E_p = I_p e^{i\phi_p}$$

$$E_s = I_s e^{i\phi_s}$$

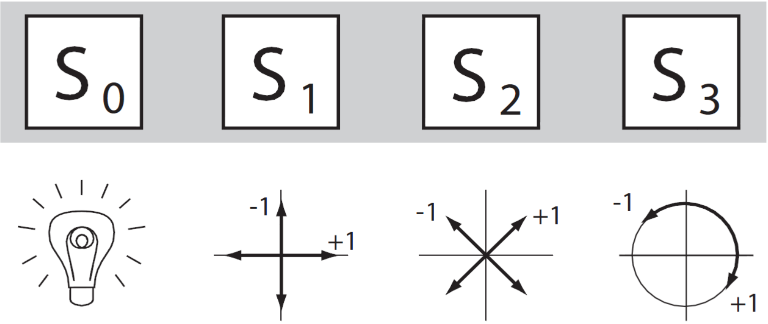
We compute the **Stokes vector** as:

$$S_0 = |I_p|^2 + |I_s|^2$$

$$S_1 = |I_p|^2 - |I_s|^2$$

$$S_2 = \mathrm{Re}(E_p E_s^*)$$

$$S_3 = \mathrm{Im}(E_p E_s^*)$$



Stokes Vectors

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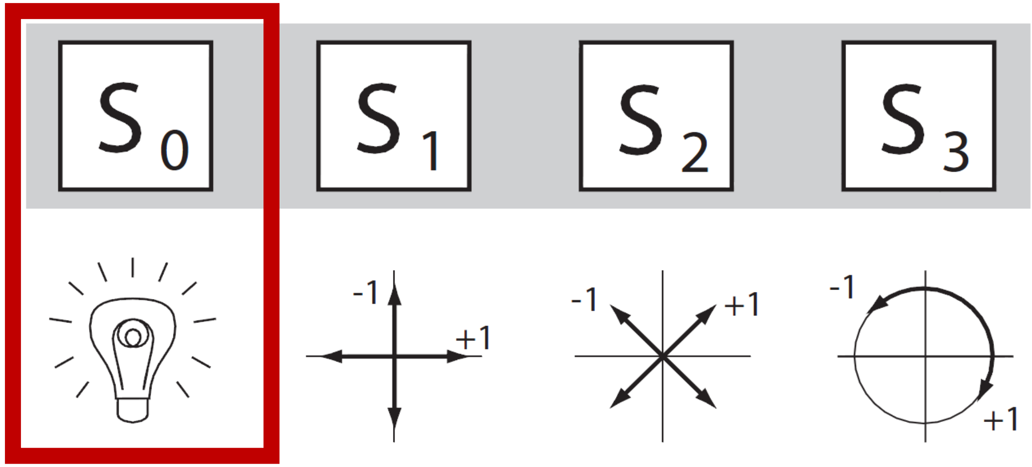


Remember:

Unfortunately, a camera only captures intensity

Polarization (Stokes) imaging

- Cameras (and eyes) do not capture polarization, just intensity
- … so the info about phase and phasor intensity is lost.

- **Polarization imaging** tries to:

  a) Recover (at least) the polarization state of light

  b) Not all info of the EM-field, but at least its macroscopic behaviour

Sources of polarization

- **Obviously, beyond polarizing filters**
- BTW, have you tried to see your phone with polarizer sunglasses?

- Reflection and Transmission in **Dielectrics**.
  - Check out the Fresnel equations
  - Metals/Conductors are weakly polarizers

Sources of polarization

- **Obviously, beyond polarizing filters**
- BTW, have you tried to see your phone with polarizer sunglasses?

- Reflection and Transmission in **Dielectrics**.
  - Check out the Fresnel equations
  - Metals/Conductors are weakly polarizers

- **Scattering** in media
  - Specially Rayleigh, but also larger particles

Why should we care about polarization?

- **Brainstorming:** Where and why would we like to have pol-info?

Removal of specular reflections

Del Molino and Muñoz 2019

Removal of specular reflections

Del Molino and Muñoz 2019

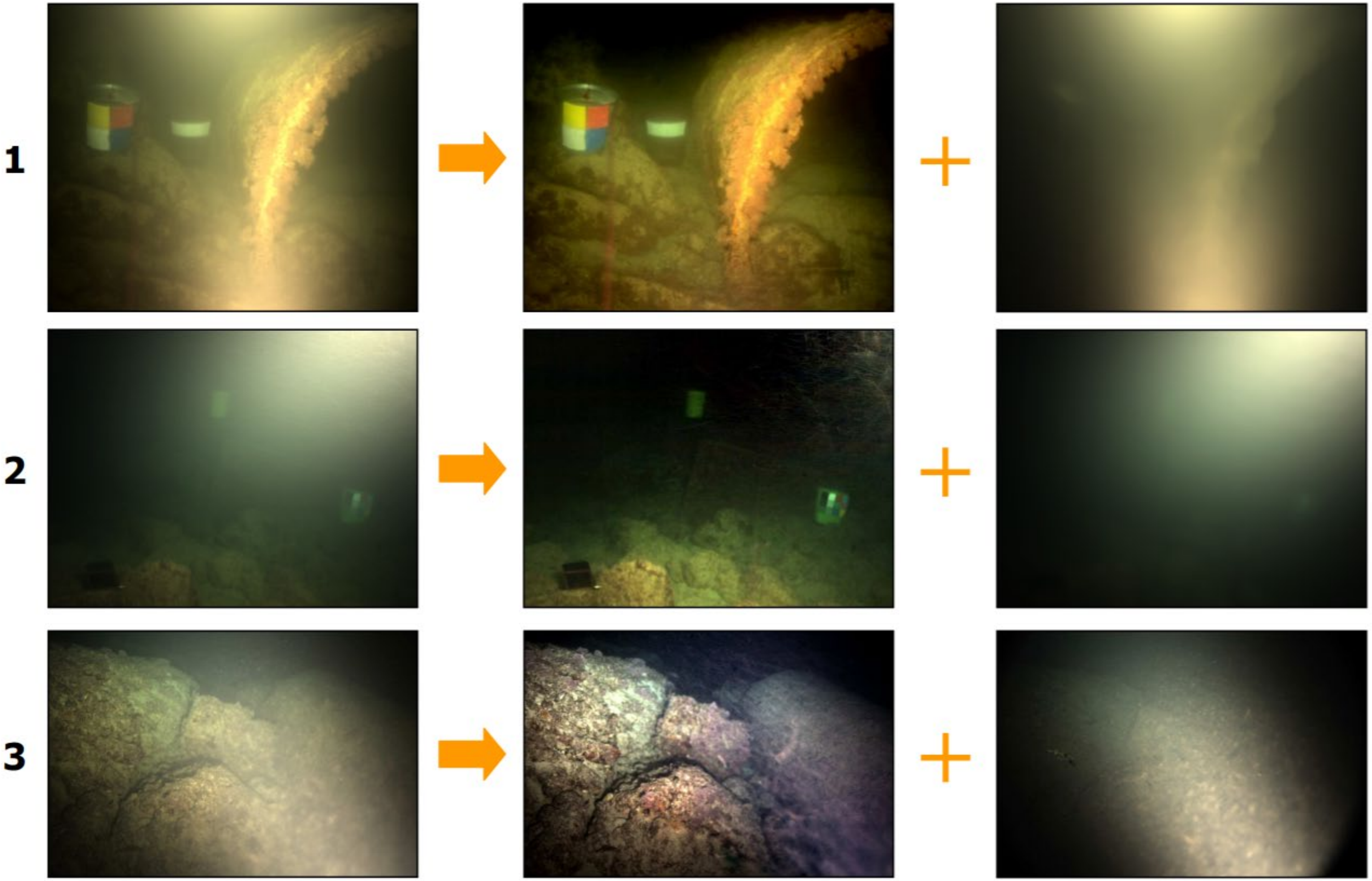
Removal of specular reflections

https://youtu.be/yhjKO1a99OQ?t=56

Specular-Diffuse separation

Ghosh et al. 2008

Vision through media - *Dehazing*



Treibitz and Scheschner 2009

Why should we care about polarization?

- **Brainstorming:** Where and why would we like to have pol-info?

- Remove specular reflections
- Diffuse/specular decomposition
- Vision through media
- Material recognition
- …

Quick Quiz

- What is the best technique for obtaining local detail in computer vision?

a) A time-of-flight camera (e.g. lidar) Transient Imaging (Lecture #9)

b) Shape-from-specular

c) Multiview-Stereo

d) Coded illumination Computational illumination (Lecture #7)

Sources: CMU 15-463, Dartmouth College CS89/189.

Why should we care about polarization?

- **Brainstorming:** Where and why would we like to have pol-info?

- Remove specular reflections
- Diffuse/specular decomposition
  - With applications on shape-from-specular
- Vision through media
- Material recognition
- …

Questions?

Polarization Imaging

- **What** is it?
- **Why** is it important? (a.k.a. applications)
- **How** do we capture it?

Polarization Imaging - *How do we capture it?*

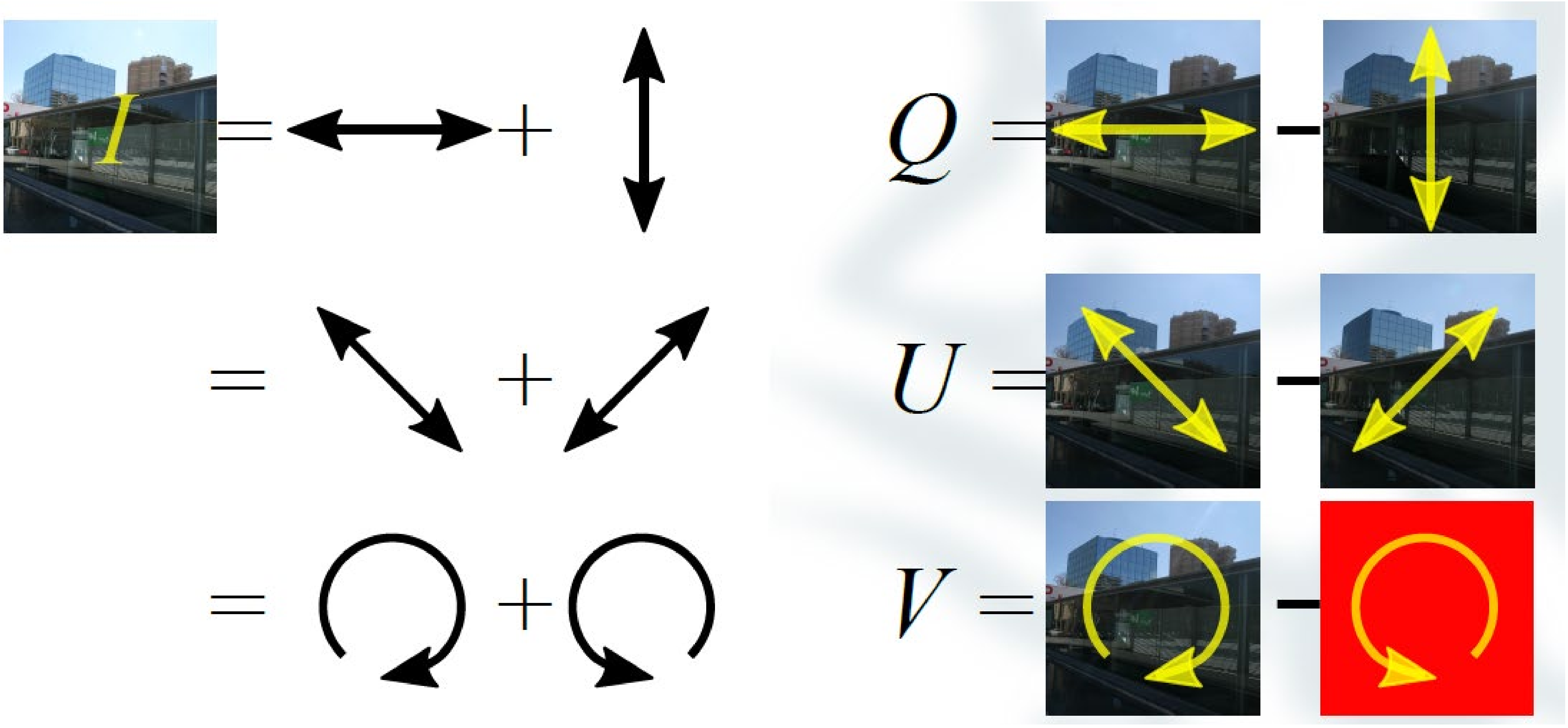
Polarization Imaging - *How do we capture it?*

- **What we need:**

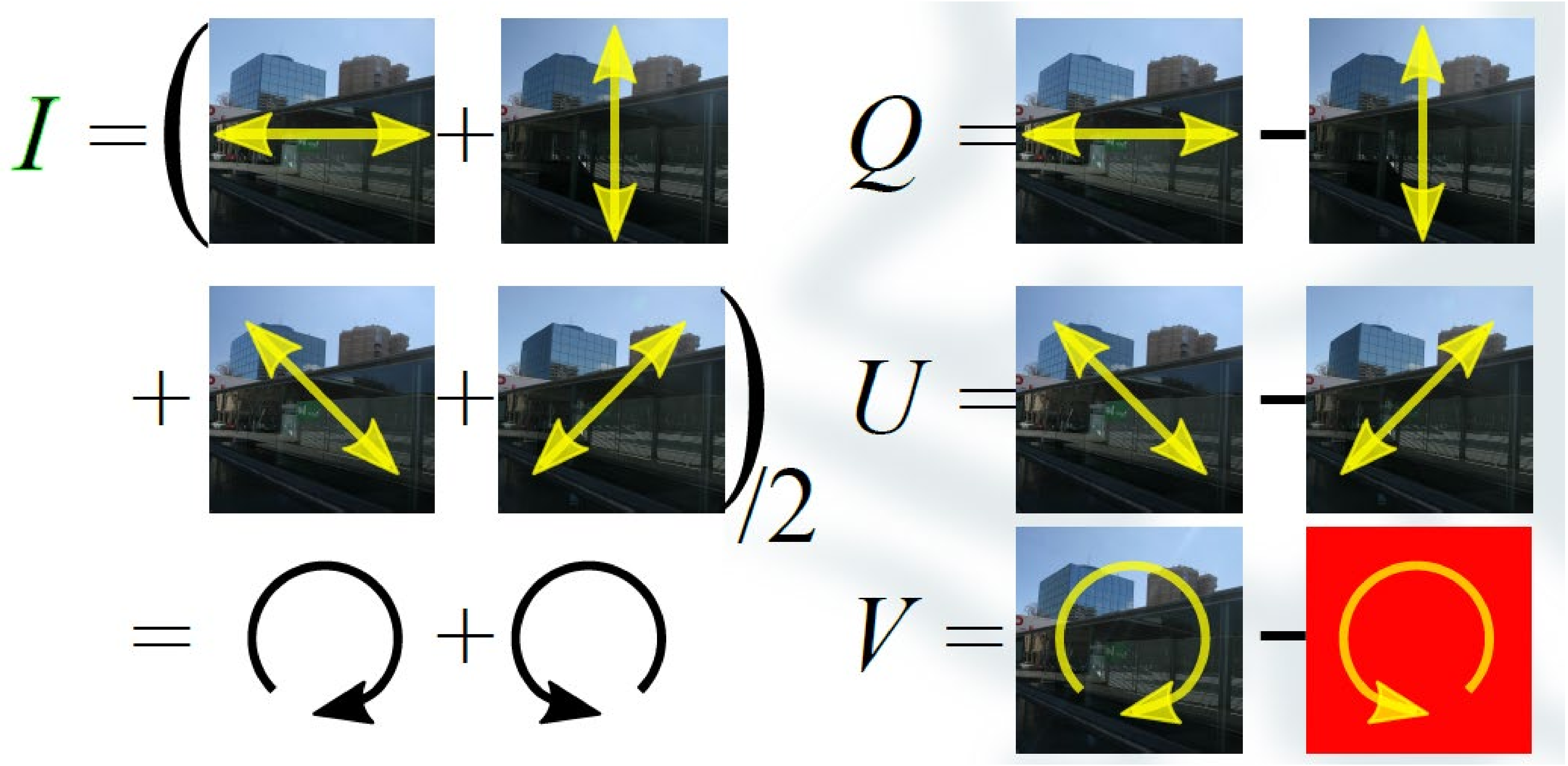
A conventional Camera

Polarizing Filters

Polarization Imaging - *How do we capture it?*

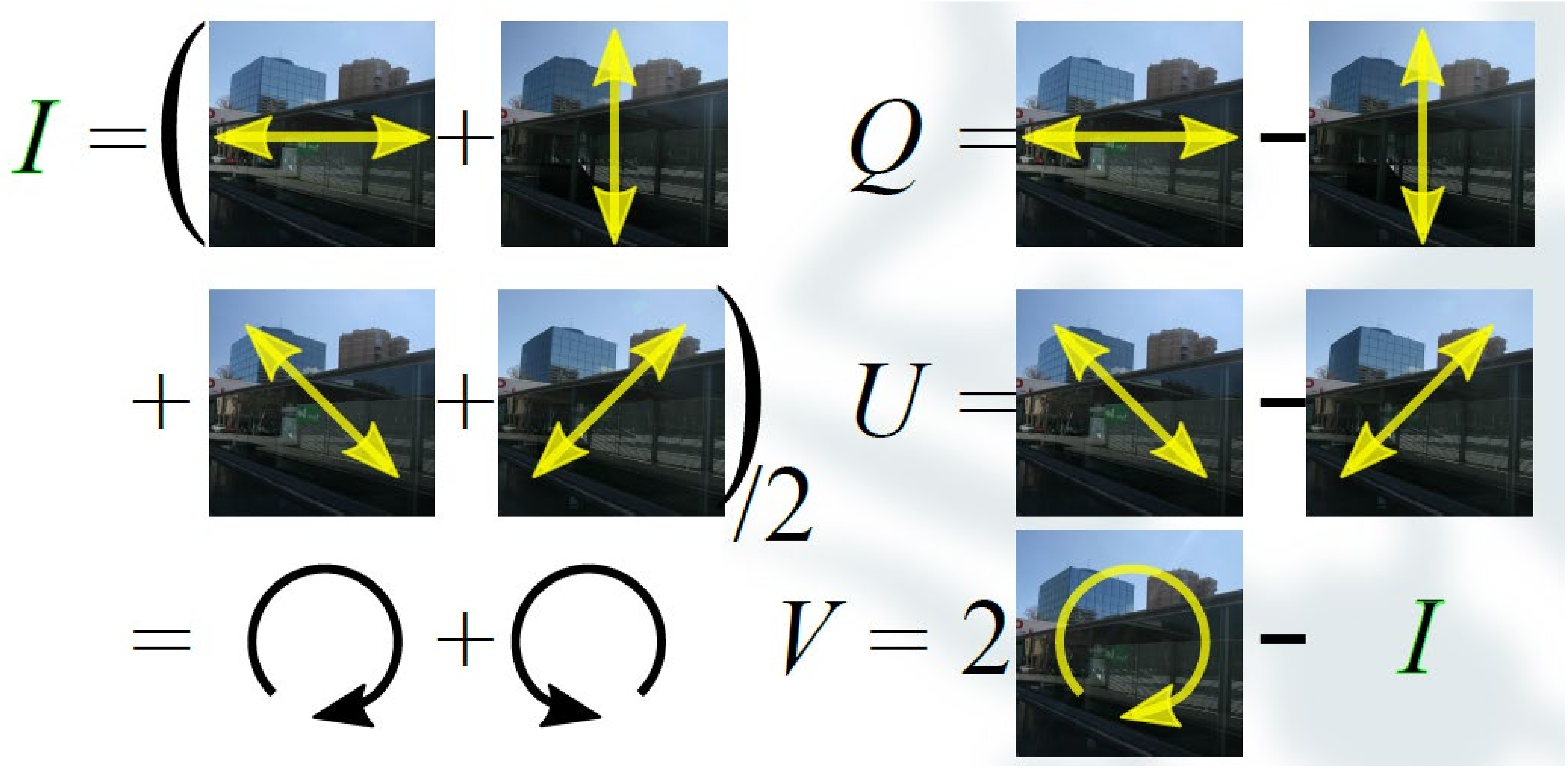


Polarization Imaging - *How do we capture it?*



Del Molino and Muñoz 2019

Polarization Imaging - *How do we capture it?*



Polarization Imaging - *How do we capture it?*

- **Stokes scanning** – *Naïve approach*
  - *Static scenes only, 5 shots (4 linear, 1 circular pol)*
- **Interferometry-based approaches**
  - *Single-shot, but super-complex setups…*
- **Some Bayer-like per-pixel filters**
  - *Expensive, only linear polarization, demosaicing hell*

Recap

- **Polarization/Stokes imaging**
  - **What:** Capturing information beyond (spectral) radiance, including phasor information of the carrier, encoded in its ***polarization*** state.
  - **Why:** Most surfaces and media polarize / depolarize light:
    - Lots of info about surfaces (optical properties) and their shape (shape from specular)
    - Removal components (specular, scattering) for improved imaging!
  - **How:** Capture the Stokes vectors, multiple measurements using linear/circular polarizing filters.

In summary – *Take home messages*

- The choice of our imaging systems is based on the **human's eye**, which is amazing, but far from perfect

- In general, we only care about **radiance** (not even spectral), neglecting the EM nature of light
  - Maybe that's why the particle hypothesis prevailed for 15 centuries…

- **Hyperspectral** and **polarization** information are very useful for scene understanding and CV. Let's use them!
  - In the end, some of the most sophisticated eyes in nature exploit them heavily.


Questions?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